

Caden Kraft — Ironless Rotor Cycloidal Planetary Actuator (및 관련 액추에이터/모터 글)

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Ironless Rotor Cycloidal Planetary Actuator

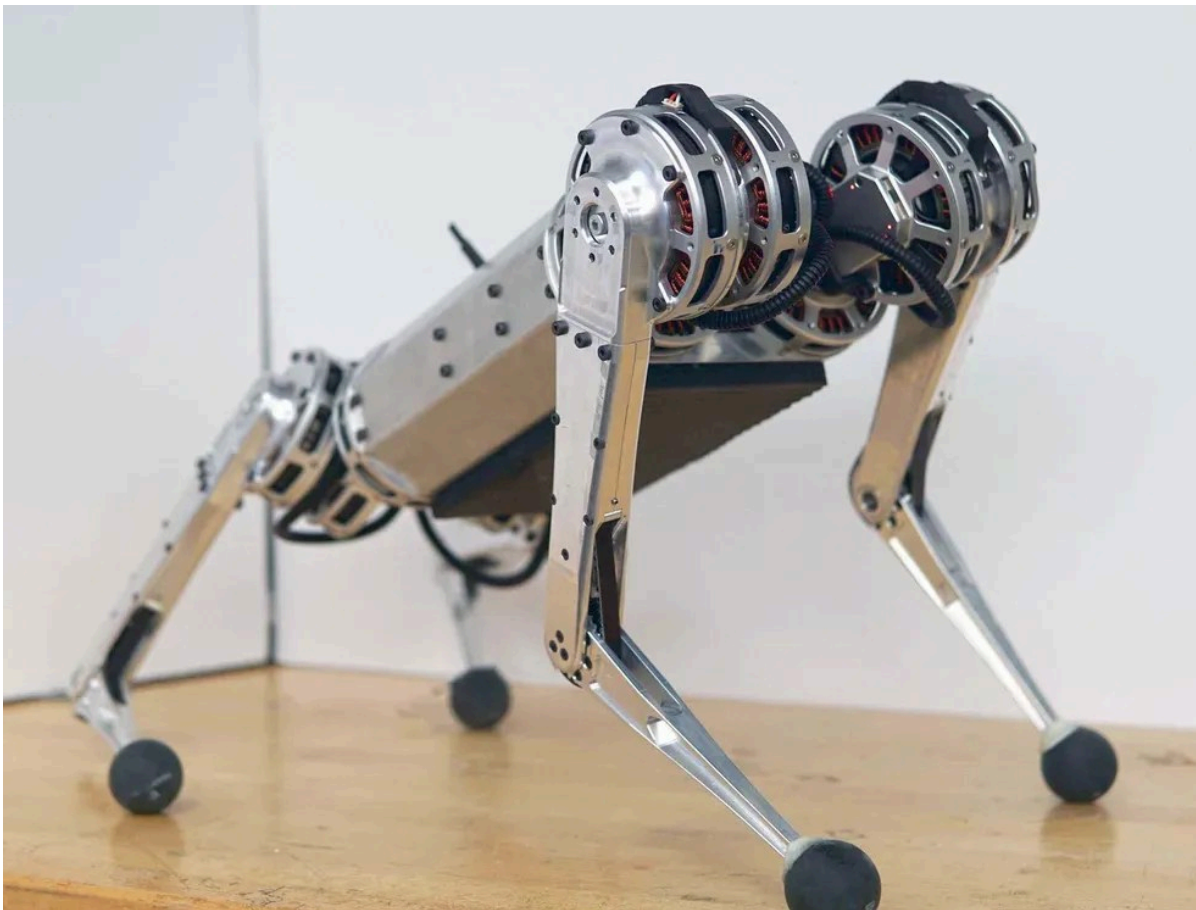
출처: <https://cadenkraft.com/ironless-cycloidal-planetary-actuator/>

In my last SCARA arm project, I had a very frustrating experience designing high torque/high accuracy gearboxes for the Nema 23 stepper motors I used. Ideally for that project I could have used fancy off the shelf servos or QDD (quasi-direct-drive) actuators like the MIT Mini Cheetah. The problem with these is that they generally cost ~\$500-1000 each, which is something I can't really justify for my projects.

This led me to want to develop my own QDD actuator with a heavy focus on keeping the cost low, both on the parts and tools needed to create one.

What's a QDD actuator?

A quasi-direct-drive actuator is a term referring to a type of actuator initially popularized by MIT with their Mini Cheetah actuator.



MIT Cheetah using 12 QDD actuators

They generally consist of a radial BLDC motor with a gearbox reduction inside the stator that can heavily increase the torque while being extremely space efficient in their height. They are most famously used inside quadruped robots like the MIT Cheetah.

For this actuator, I don't have a final plan on what I am going to use it on just yet, but I wanted to create a platform that I could easily use and modify for my future projects.

Project Goals

The main goals I had for this project are as follows:

- Low cost, less than \$80 per actuator
- Fully 3D printed, every single custom part is 3D printable. No machining, laser-cut metal parts, etc
- 10Nm of holding torque
- Easy to interface into my projects

Some of these goals present some challenges, namely needing to make the rotor and gearbox fully 3D printed.

Rotor Design

Every actuator design I have found online uses some sort of iron/steel backing behind the magnets in order to direct the flux of the magnets inside towards the stator. If my rotor is going to be made out of plastic, I wanted to try and make a design without a ferrous backing that doesn't give up significant performance.

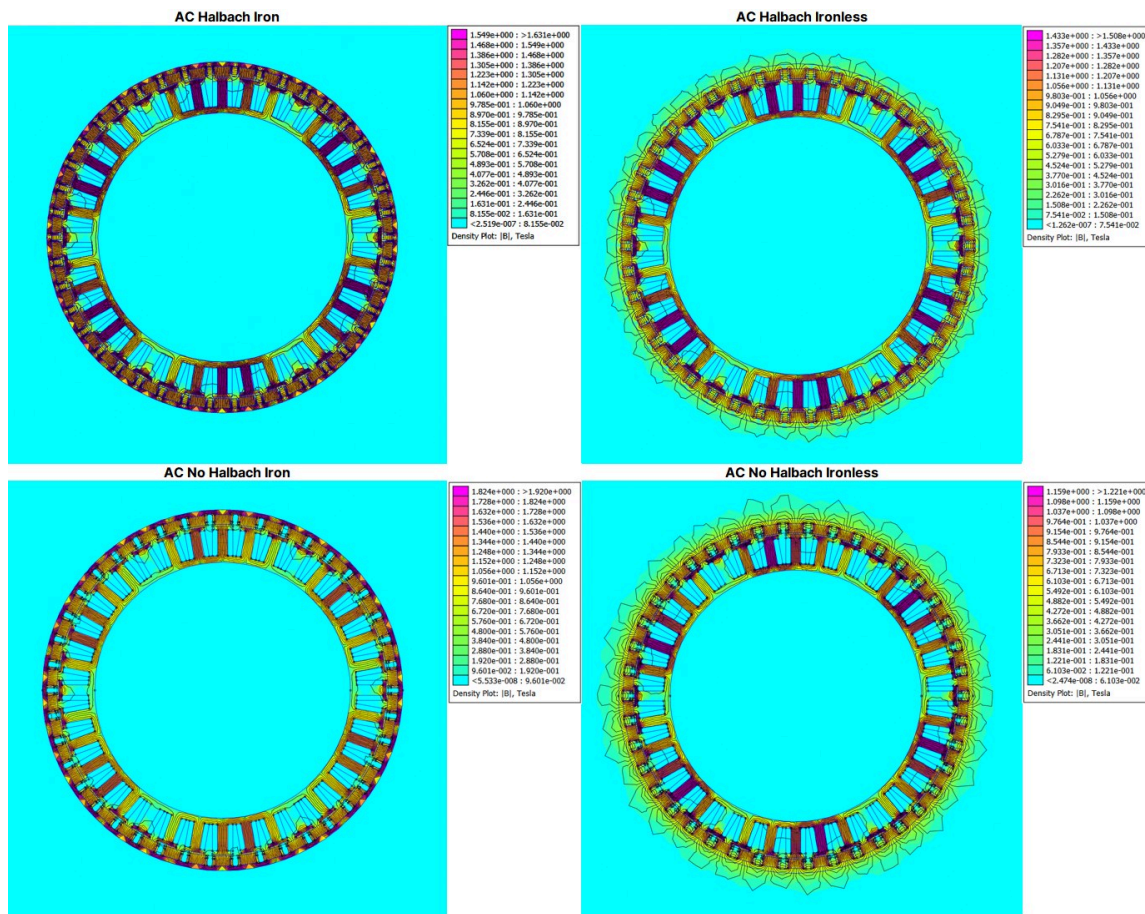
To solve this, I chose to use a Halbach array like my previous axial flux motor project that was also fully 3D printed. This time I actually wanted to quantify the performance that I was leaving on the table by not using an iron backing.

I used FEMM to create magnetic simulations where I measured the static torque at a fixed angle for all three phase pairs of the motor. Then I created 4 different configurations to test how the different rotor designs affected performance: Halbach array with an iron backing, Halbach array with no iron, normal array with iron, and normal array without iron.

The results of these simulations are below.

Rotor Config	AB Torque (Nm)	BC Torque (Nm)	AC Torque (Nm)	Average Torque (Nm)	% From Max
Halbach Iron	-0.915	-0.310	1.824	1.017	100.000%
Halbach Ironless	-0.832	-0.207	1.748	0.929	91.381%
No Halbach Iron	-0.953	-0.293	1.540	0.928	91.328%
No Halbach Ironless	-0.791	-0.243	1.175	0.737	72.455%

Note that the torque values here are not really representative of the actual torque of the actuator and just used as a point of comparison.



Density plots from FEMM output

These results were very interesting to me. Not only did the Halbach array have nearly identical performance to an iron-backed rotor, using both gets you a 9% bump in torque, which I would not have expected. This also goes to show how much performance I would be leaving on the table if I just used plastic parts without a Halbach array.

I also want to stress that I don't have a ton of experience in magnetism so if anyone has feedback on these simulations or tips for what I could change I would love feedback!

Using a Halbach array instead of an iron backing will also have much lower rotational inertia compared to an iron backing which is important for an actuator designed to quickly change directions.

Below is the final rotor design I came up with as a result of these simulations. As you can see, the magnet array is very dense with 42 12 x 5 x 3mm N52 magnets for the main pole pairs, and 42 1/2 x 1/8 x 1/16 N50 magnets for the Halbach pairs, both of which have a 0.7mm air gap to the stator.

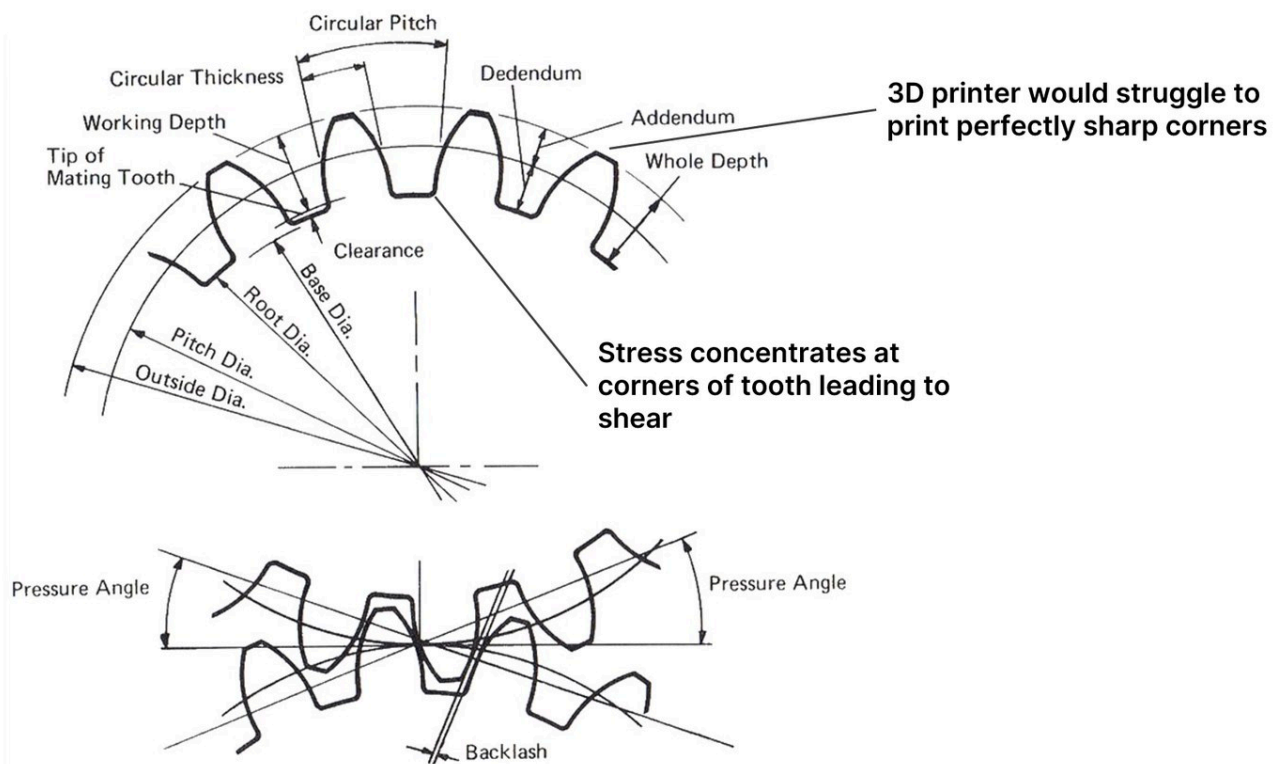


Render of final rotor design

Gearbox Design

The next big hurdle I had to overcome was designing a gearbox capable of handling the high expected torques of this actuator made out of 3D printed plastic parts. I chose to go with a planetary gearbox as it can usually be made more compact as compared to cycloidal or harmonic gearboxes.

The problem with 3D printing gear teeth is that most gear tooth profiles require sharp edges and because 3D printers extrude plastic out of a round nozzle, you are always going to have some sort of corner radius no matter how small. Plastic gears using traditional spur gear teeth also tend to run into shearing issues as the teeth are quite small and tend to fail at stress concentrations near the base of the tooth.

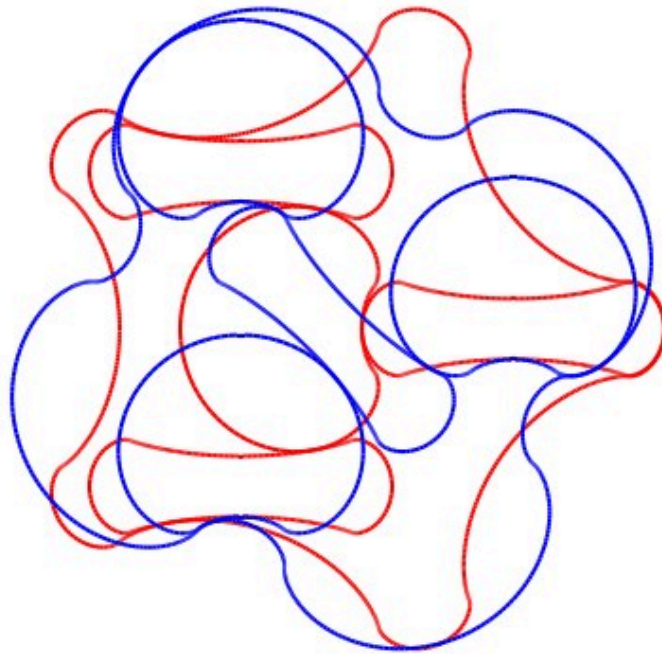


Traditional spur gear tooth profile

This made me rethink how I approached defining my tooth profile. The work I did on my previous cycloidal drive projects made me wonder if there existed planetary gearboxes that used cycloidal 'tooth' profiles.

Searching around I found out about pygeartrain, a Github project that can generate all sorts of crazy gearbox designs from mathematical functions. This can lead to some really wacky planetary designs such as this one. All credit for this goes to Eelco Hoogendoorn and he has a ton of other cool projects!

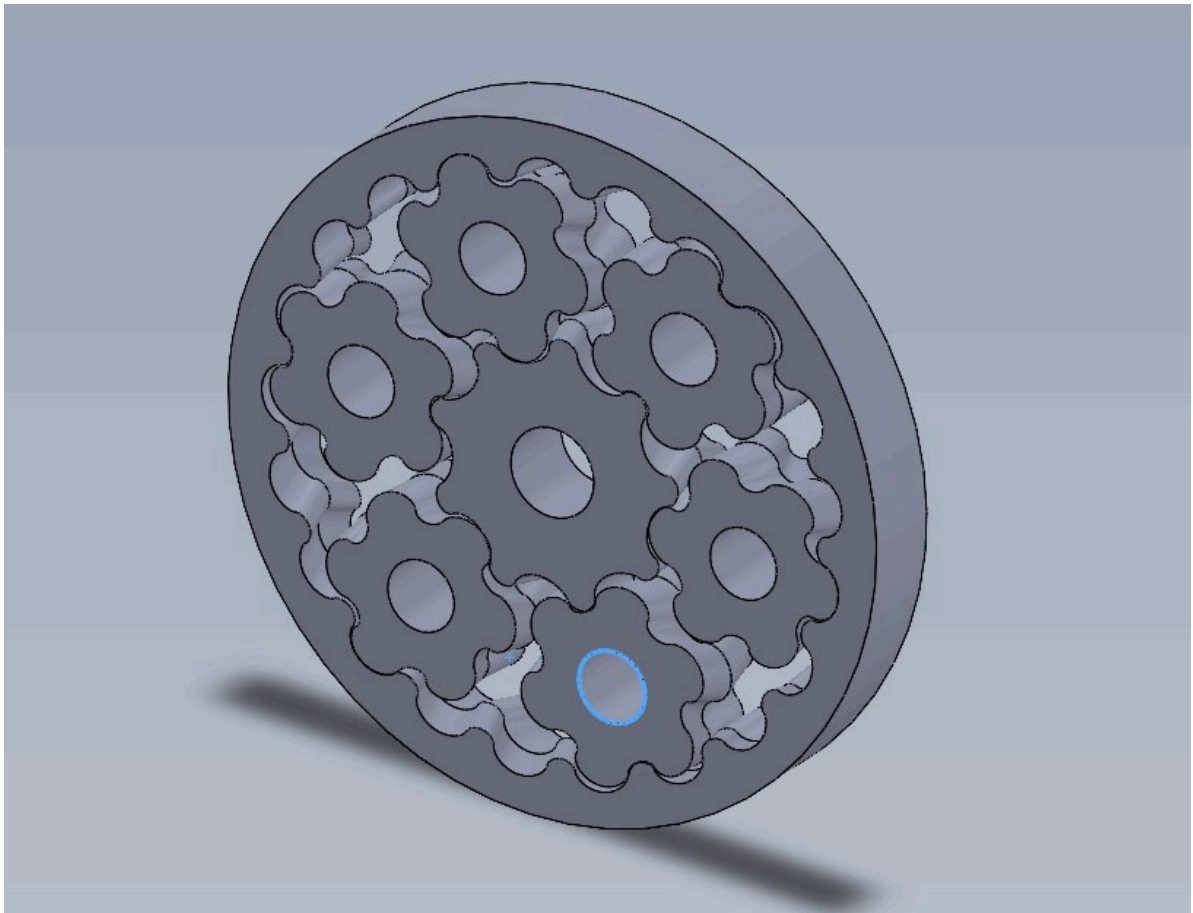
s1/r2: 16 + {R1:5,P1:2,S1:1,R2:4,P2:1,S2:2}



pygeartrain output for lowest known tooth count compound planetary

The problem is that pygeartrain was designed simply as a way to test and visualize gear profiles and not to create anything you could actually export to a 2D/3D model and implement into a design.

This led me to create a version that can create the cycloidal planetary gearbox profiles, allow you to give them actual dimensions, as well as create export files for actually modeling them out in SolidWorks. I also added functionality for creating helix and double helix versions of the cycloidal gear tooth profile.



Double helix cycloidal tooth profile exported using my script

Below is my Github repo where you can mess around creating your own wacky gear profiles with a full guide to turn those into 3D models.

GitHub - CKraft11/pygeartrain: Gearing calculations and related functionality

Gearing calculations and related functionality. Contribute to CKraft11/pygeartrain development by creating an account on GitHub.

GitHub

CKraft11

CKraft11/ pygeartrain



Gearing calculations and related functionality

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Contributors

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Issues

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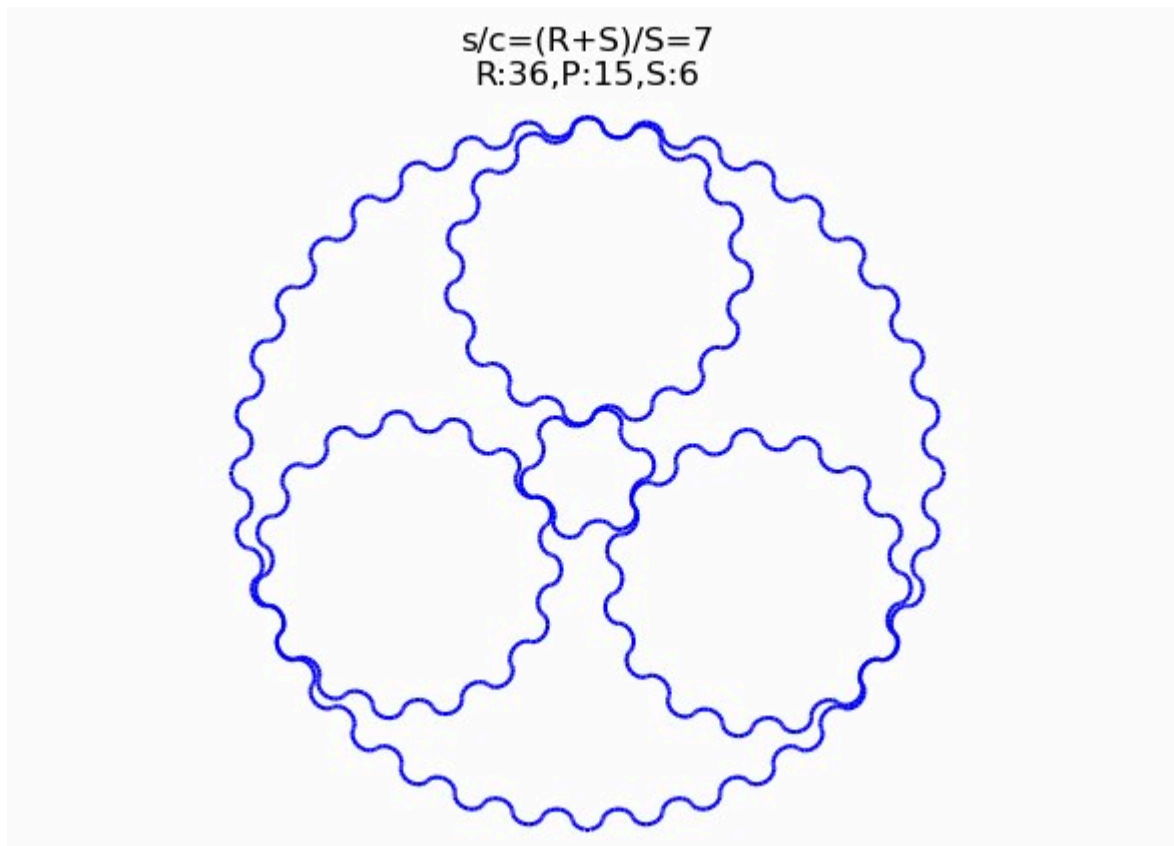
Stars

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Forks



For this project I chose to create a 7 to 1 gearbox visualized below.



pygeartrain output with of chosen gearbox

As you can see with this gear profile, not only is there constant contact with no backlash due to the profile, you can also see the "teeth" are more like lobes in one continuous outline, which a 3D printer can easily create. Both of these benefits make these gears seem like they should be fantastic for 3D printed gearboxes, and I am honestly surprised I haven't seen anyone try this before.

Creating the integrated design into my actuator from these exports was slightly more complicated. Below is the final gearbox module that goes through the center of the actuator.



Gearbox module from the actuator



Exploded view of gearbox module

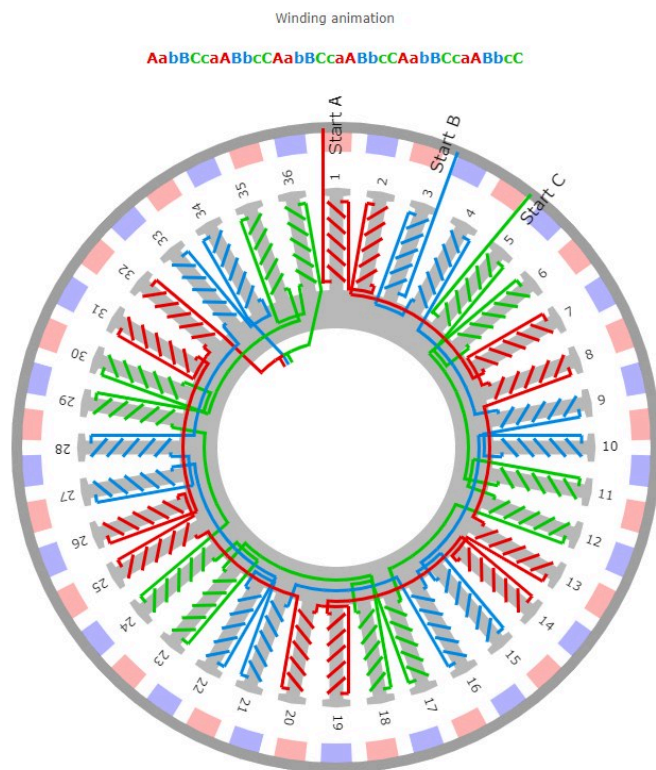
Stator and Windings

For the stator, I chose to use a 10010 stator as they are cheaply available online and fell roughly within the size I wanted the stator to be. This is a 36 slot stator so I chose to use this winding configuration that I got from the bavaria direct winding calculator. For the windings, I chose to do 6 turns (5 full, 1 partial) around each slot, winding 6 strands of 0.4mm diameter wire in parallel.

Slots
 Magnet Poles
 2 layer

 = advanced

This slot/magnet pole combination will have 252 cogging steps per turn. $LCM(36,42)$
 And its winding factor is: 0.93301 advanced



Here is a render showing how the stator is assembled into the clam-shell of the gearbox module.



Render of gearbox module with stator

The Final Actuator Design

Now that I had designed a rotor, gearbox, and chosen what stator I wanted to use, it was time to put everything together into one cohesive design. I wanted this actuator to have an integrated controller and to keep costs down I chose to go with the MKS X Drive Mini. This is a low cost controller based on the older ODrive 3.6 and has a built in hall effect encoder making it suitable for integrating directly into the actuator.

I would have liked to use a proper new ODrive or a moteus controller, but they are hard to justify as they cost 2-5x more than my entire actuator design for just the controller.

Here are some renders of the final actuator design that I came up with.

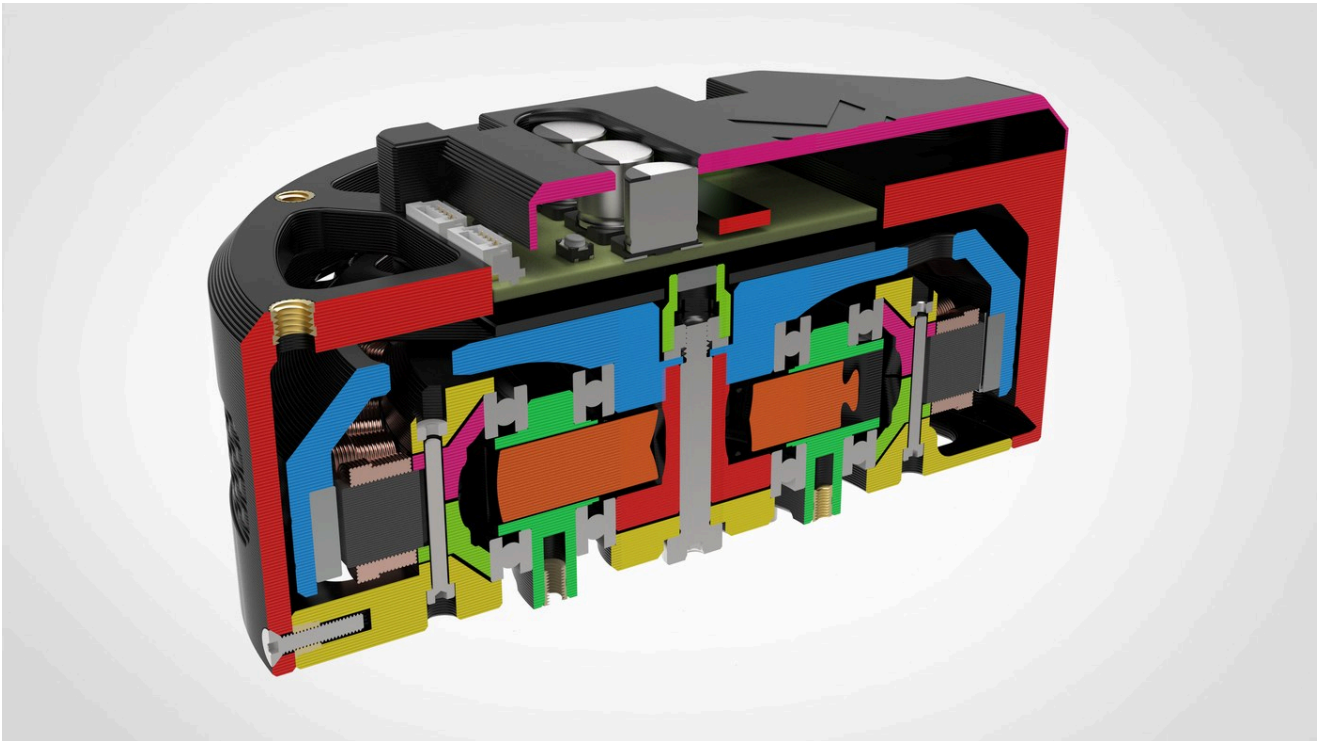


Render of final design



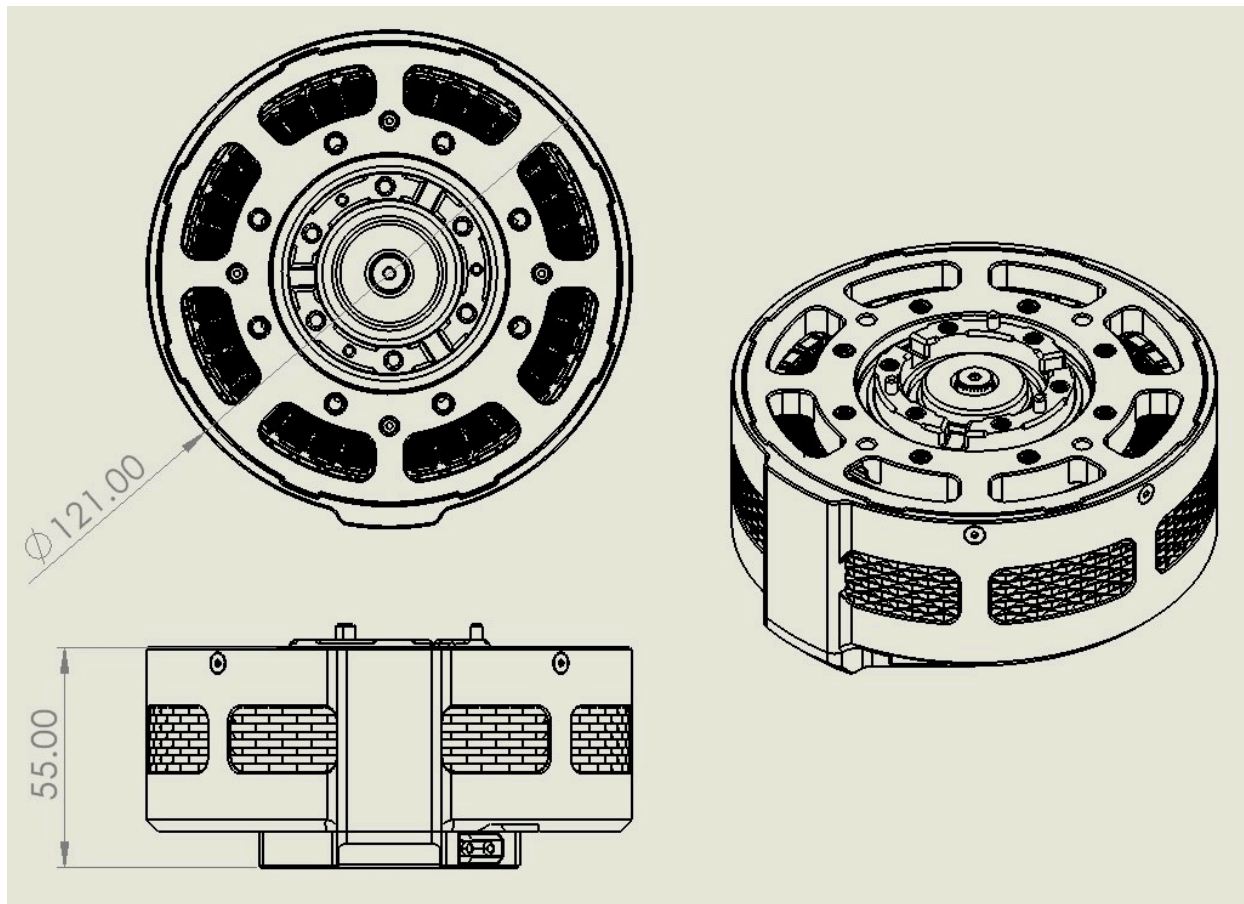
Exploded view of entire actuator

Here is a cross section of the actuator with all the 3D printed components colored.



Section of the actuator with all the plastics highlighted

And here are the final bounding dimensions of the actuator.



Fabrication and Assembly

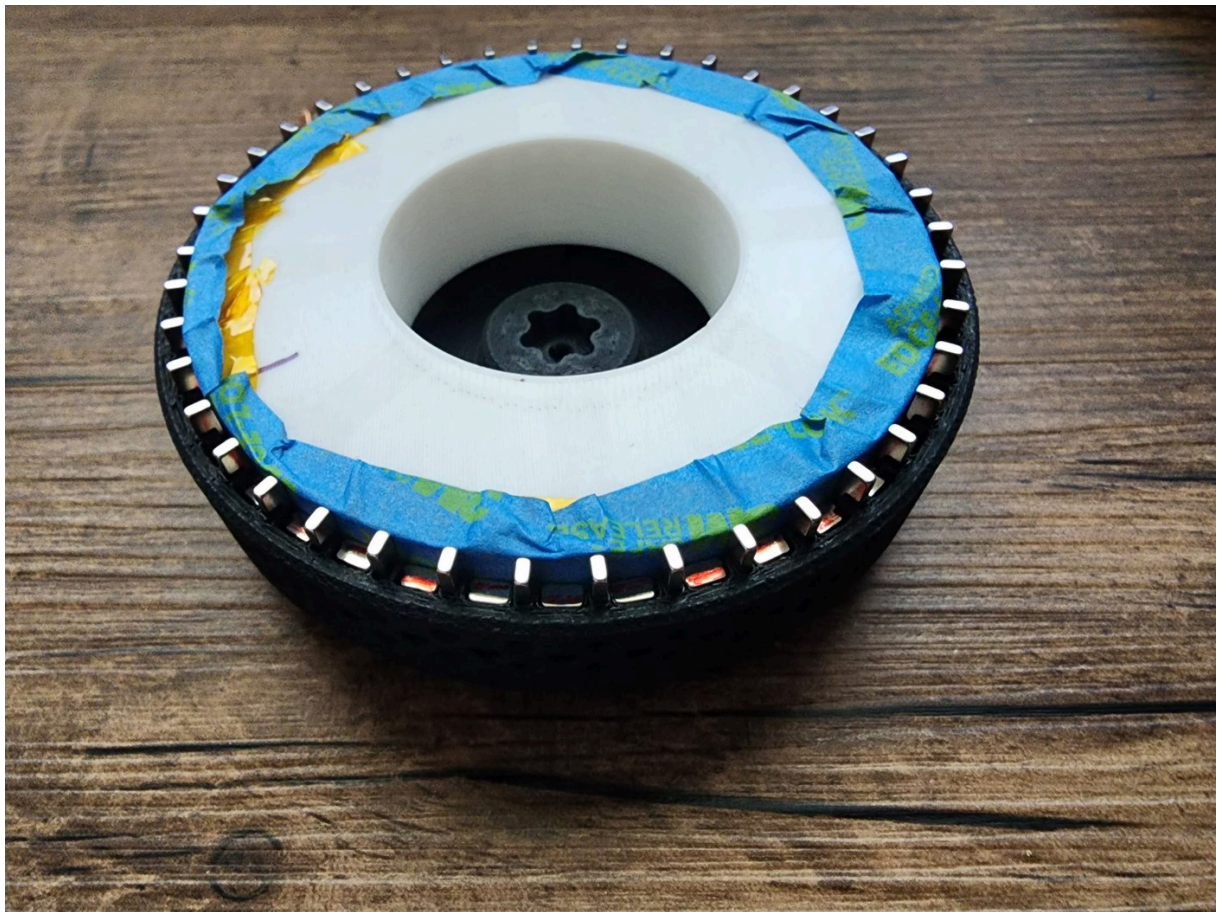
With all the custom parts on the actuator being 3D printed, most of the motor fabrication was just printing the components out. I printed all of the parts out of PA6-GF for some extra strength but I probably should have used just normal nylon filament for the gears if I had to do this again because the glass additive might cause abrasion over time.

I did go through quite a few prototype parts making this though as I went through several different designs for nearly every part.



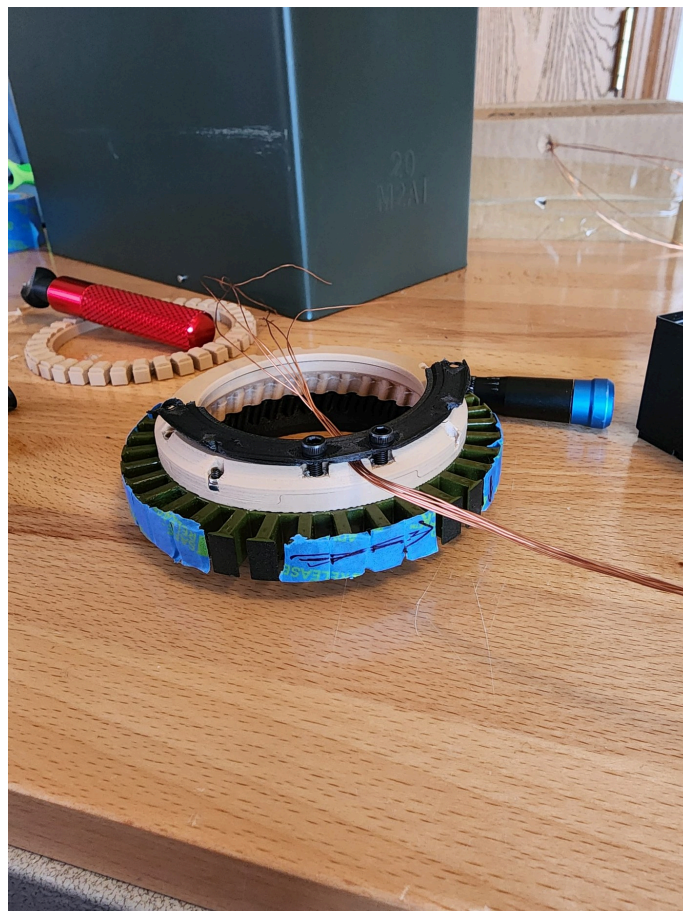
All the failed/revise 3D printed parts for this project

When assembling the rotor, I created a fixture to hold all my magnets at the desired air gap and then slid them in from the bottom. There are radial holes behind every magnet that I used to superglue them in place from behind after all the magnets were slid in place.



Finished rotor (left), Magnet insertion fixture (right)

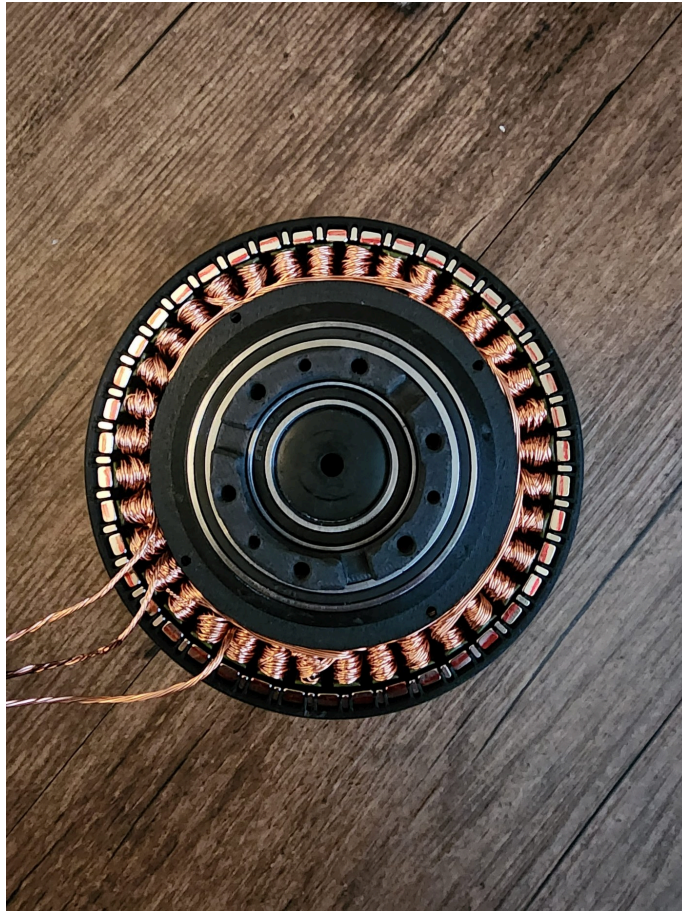
For the winding, it was a bit more complicated. I started by using the incredibly advanced tooling seen below to split 1 large spool into 6 smaller ones and then I fed all 6 through an equally sophisticated enclosure to get ready for winding.

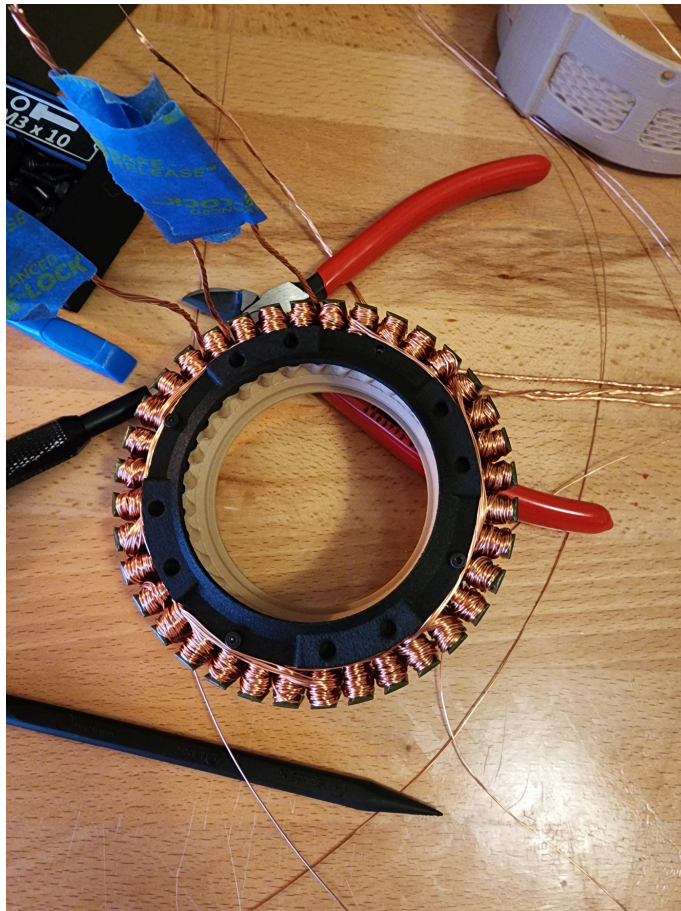




Photos for how I set up each individual copper spool and fed them all together to wind the stator

I then completed the arduous task of winding each phase, which in total took about 3 hours. After that, I took the wound stator to an LCR meter at my university in order to check the phase resistances and see if I had made any winding mistakes.





Finished wound stator as well as testing on the LCR meter

The results from the LCR meter are below. As you can see the tolerance between coils is incredibly good and far better than any of my other motors.

	Resistance (mΩ)	Impedance (μH)
φA	62.36	33.92
φB	62.42	33.04
φC	63.31	33.92
Tolerance	±0.76%	±1.31%

After assembling the gearbox with some lithium grease for lubrication I completed the assembly of the actuator.

Here is a video where you can see the gearbox working.

▶ 임베드: <https://www.youtube.com/embed/x7s6tjZYdYg?feature=oembed>



Final assembly (left), gearbox (right)

The final actuator has zero backlash, has little to no resistance when back-driven, and weighs in at 728 grams.

Testing

To start off testing this actuator I printed a 250mm arm that I could swing around and use to have a nice visual when calibrating the PID values for the FOC controller.

Here is a video of one of the first times I actually got it moving.

▶ 임베드: https://www.youtube.com/embed/KsSsMt_DJR4?feature=oembed

I also needed to figure out the torque constant and Kv for further tuning of the actuator. To do this, I backdrove the actuator with a power drill at a known RPM and measured the voltage generated from one of the phases.

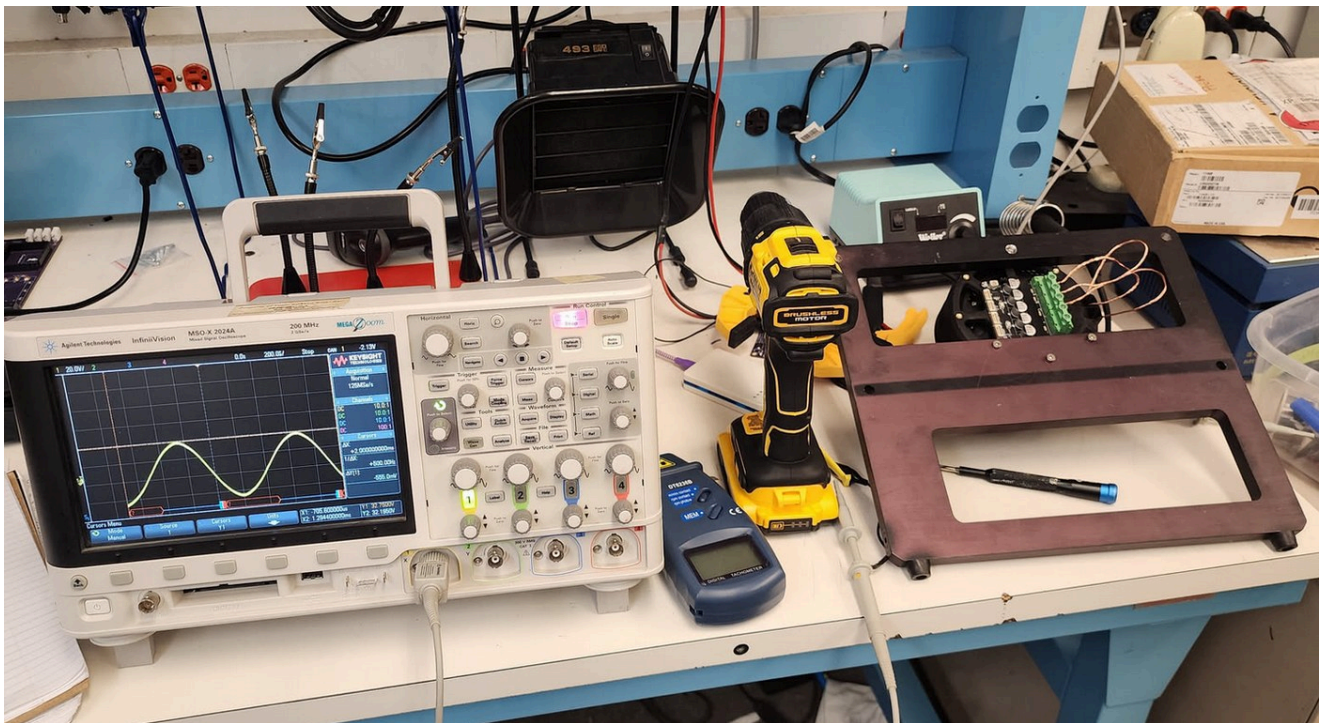


Photo of the testing for finding the Kv

From this, I calculated that my actuator has a Kv of ~79, which lines up with what I would expect and allowed me to calculate the torque constant for the ODrive config.

After I got it nicely tuned, I set up my lab scale below the end of the arm and did a simple test to see what the holding torque would look like.

▶ 임베드: <https://www.youtube.com/embed/1illuCFuXWY?feature=oembed>

In this test the motor was able to hit a peak holding torque of 14Nm which was decently above my goal of 10. However, even with it outperforming it still looked like there was room to go further. The actuator only stopped because it had reached the current limit of my 24V 25A Meanwell power supply.

That meant I needed to find a DC power source that could push a lot of current. After asking around some forums for a power source people recommended for a project like this, I was pointed in the direction of reclaimed battery modules out of EVs. Luckily for me, after seeing my messages, Peter

Holderith (@_baldtires) said that he had an extra module from an Audi E-tron that he didn't need and would send me for free.



13S Audi E-tron module I used for testing

This was extremely helpful and was the only way I would have been able to complete this project, so a big thanks to him. Check out his cool projects at the link above.

The module is able to be charged to 50V and has a maximum peak current output of 400A. 20kW of power was more than enough for what I would need during my next test, so I created a new setup powering the actuator off the Audi module.

For for this next, test the actuator had well exceeded the limits of my lab scale, so I hooked a string to a force gauge and pulled the actuator perpendicularly to the arm.



Second test setup powered off the Audi module

I also set up thermocouples inside the coils in order to see when I would be getting close to melting the insulation.

With this setup I was able to reach a maximum holding force of 117.5N which equates to a holding torque of 29.4Nm with the wires reaching a surface temperature of 78°C. Even in this test there was no mechanical or electrical failure and it only stopped because the controller had reached the 50A current limit I had set. However, I got the feeling that I was reaching the edge of what both the coils and plastic gearbox were capable of, so I left it at that. This gives the actuator a respectable torque to weight ratio of 40.38Nm/kg.

One crazy thing is that after a test of the actuator holding ~30Nm of torque for several minutes, there was still no perceptible backlash in the gearbox.

Cost Breakdown and Building Your Own

Of all the stuff I have created on this website I think this project is definitely something that people could benefit from and use in their own projects. This was the driving reason why I wanted to make this as low-cost as possible. The final bill of materials breakdown is below.

QDD Actuator BOM					Pricing Calculated 4/22/25	
Part Name	Part Link	Part Quantity	Cost Per (Part/Gram)	Total Part Cost		Total Cost w/ Controller
6805 Bearing	https://www.amazon.c	2	\$ 1.29	\$ 2.58	\$	72.63
6710 Bearing	https://www.amazon.c	2	\$ 2.26	\$ 4.52		
Makerbase MKS XRIVE Mini	https://www.aliexpress	1	\$ 29.99	\$ 29.99	\$	42.64
N52 12x5x3mm	https://www.aliexpress	42	\$ 0.12	\$ 5.00		
N50 1/2 x 1/8 x 1/16in	https://www.magnet4s	42	\$ 0.14	\$ 5.88		
0.4mm Enameled Magnet Wire	https://www.aliexpress	60	\$ 0.02	\$ 1.20		
10010 Stator	https://www.aliexpress	1	\$ 17.63	\$ 17.63		
5x12mm Pin	https://www.aliexpress	3	\$ 0.28	\$ 0.85		
M2.5x12mm Countersunk	https://www.aliexpress	20	\$ 0.03	\$ 0.62		
M2.5xL4xOD3.5	https://www.aliexpress	7	\$ 0.03	\$ 0.20		
M3xL4xOD4.2 Brass Insert	https://www.aliexpress	14	\$ 0.03	\$ 0.45		
M4xL4xOD5 Brass Insert	https://www.aliexpress	15	\$ 0.03	\$ 0.52		
M4x25mm Shoulder Bolt	https://www.aliexpress	1	\$ 0.53	\$ 0.53		
M2 Locknut	https://www.aliexpress	4	\$ 0.06	\$ 0.24		
M3x6mm	https://www.aliexpress	8	\$ 0.04	\$ 0.32		
M5 Locknut	https://www.aliexpress	1	\$ 0.06	\$ 0.06		
M2x25mm	https://www.aliexpress	4	\$ 0.03	\$ 0.13		
5x14x5mm Bearing	https://www.aliexpress	6	\$ 0.32	\$ 1.92		

Actuator bill of materials (linked in the Github repo)

I ordered these parts before the US tariffs went into effect, so pricing and availability may vary.

As you can see the final cost per actuator is just around \$40 for the actuator and 70 if you include the controller.

This gives the actuator a cost to torque ratio of \$2.47/Nm.

This is one of, if not the cheapest, actuator I have found online that has this level of performance.

All the motor CAD, FEMM simulations, BOM, and the ODrive config are open source and found via the link below. I only ask that If you use these files for your project that I be credited.

GitHub - CKraft11/Ironless-QDD-Actuator: Low-Cost Ironless QDD 3D Printed Actuator

Low-Cost Ironless QDD 3D Printed Actuator. Contribute to CKraft11/Ironless-QDD-Actuator development by creating an account on GitHub.

GitHub

CKraft11

CKraft11/Ironless-QDD-Actuator



Low-Cost Ironless QDD 3D Printed Actuator

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Contributor

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Issues

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Stars

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Forks



If you're just looking to download and print the actuator, I've also linked the print and CAD files on MakerWorld.

Conclusion

Out of all my projects this one was by far the one I had the most fun making. A lot of my other projects have mostly been useful for me to learn different things, but there wasn't much application in the final product. For this one however, I think I was able to implement some very novel ideas into a cost-effective end product that solves a real problem that I and hopefully others have. This project very much feels like the culmination of all my other motor projects, and I am really excited to see how people build off it!

3D Printed Dual Cycloidal Actuator

출처: <https://cadenkraft.com/3d-printed-dual-cycloidal-actuator/>

This project is a continuation of my previous axial flux motor project. If you've not yet read that I would recommend doing so.

Ironless Axial Flux Motor

After my first custom brushless motor I felt confident enough with the basic theory of how they work to make a second design with significantly more thought and math defining the design. Creating a 3D Printed Brushless Motor With my work in the Iowa State Solar Car club I have learned

Caden Kraft

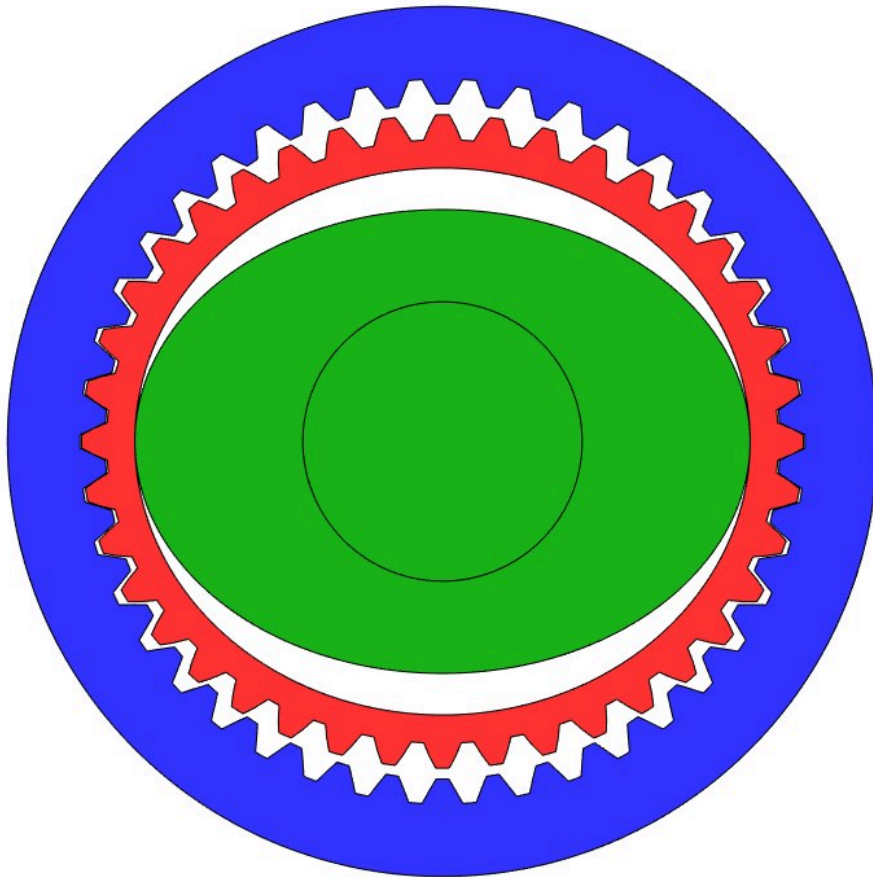
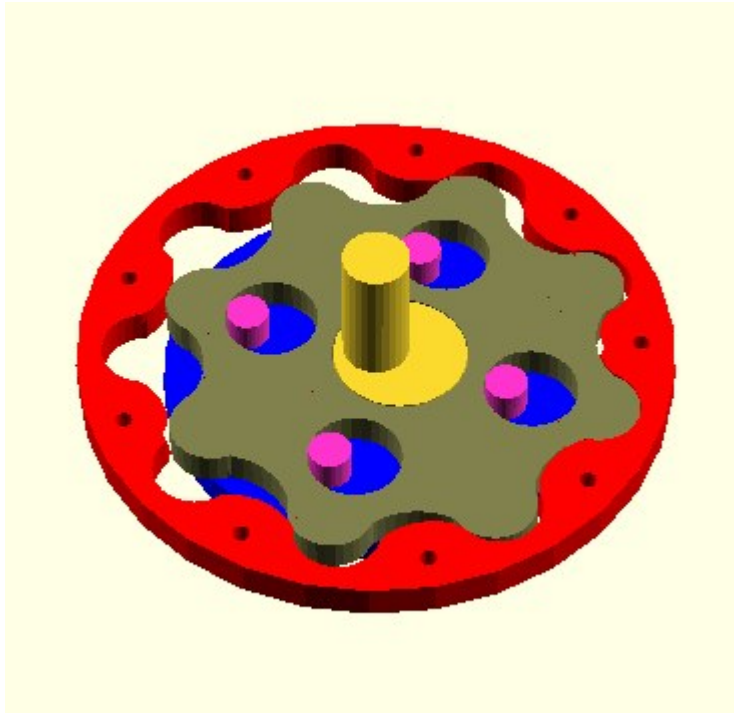


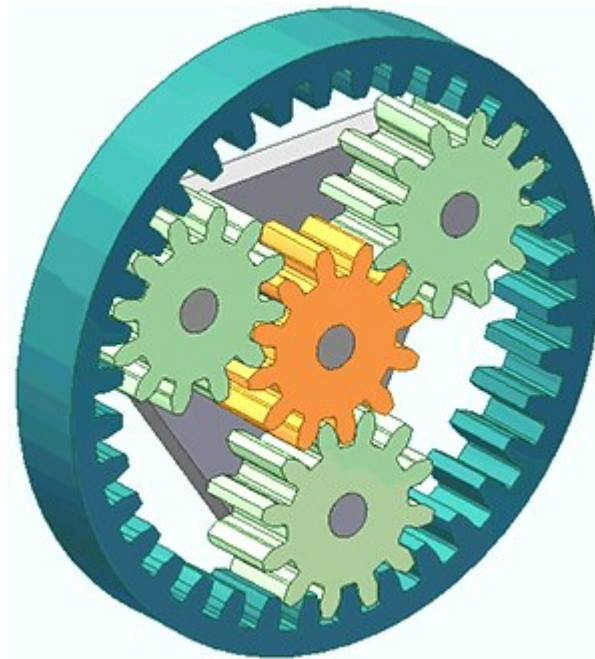
After I was finished with making my last motor project I was quite proud of the results. However, while the motor could achieve extremely high RPM's, this came at the cost of torque.

Originally I thought about rewinding the stator and possible improvements via stronger magnets and/or Iron based 3D printer filament for the stator coils among other things.

Deciding on a Gearbox

Wanting to work on something a little more different than pure motor design and not wanting to rewind any coils, I set my sights on developing a gearbox that could be integrated into my current stator design. There were 3 primary options that I was looking into when deciding what gearbox that I was going to make, Planetary, Cycloidal, and Harmonic.





Cycloidal (left), Harmonic (center), Planetary (right)

All of these options definitely seem like good ideas but I had to narrow down which one I was going to choose.

Planetary would have been the easiest and probably would have performed the best but I cut it out because I didn't really find it interesting enough and wanted to learn more about the other gearboxes.

While I admire the simplicity of the harmonic drive I was wary of my ability to manufacture the flexible parts needed. I eventually chose the cycloidal drive because it seemed like the biggest technical challenge to design as well as the fact that it looks extremely cool. Cycloidal gearboxes also have nearly no backlash which definitely made it stand out against something like a planetary gearbox with the tolerances of 3D printed parts.

Defining Project Requirements

Now that I knew what gearbox I wanted to make I started to define what I wanted the final design to look like.

- Dual cycloidal drive to reduce vibrations
- Integration of motor and gearbox into one assembly
- 70mm Z height restriction to keep a low profile
- 10:1 reduction
- Easy assembly/disassembly (no adhesives)
- Use existing stator design from axial flux motor

Actuator Design

With my requirements set, I moved on to designing the actuator. The base motor design is based on my last axial flux motor so I began modifying that in order for it to fit a gearbox. Next I had to start work on the design of the cycloidal gearbox system itself. This was quite challenging to figure out initially because the cycloid geometry is fully defined off of parametric equations.

This guide was immensely helpful in defining the geometry. After some time I had the parameters for my parametric equations ready to go based on the size and reduction that I wanted.

Parameter	Variable	Value
Radius of rotor	r	67mm
Radius of rollers	rr	4mm
Eccentricity	e	3mm
Number of rollers	n	10

$$\begin{aligned}
 x &= r \cos(t - rr \cos(t + \psi) - e \cos(nt)) \\
 &\quad + r \sin(t - rr \cos(t + \psi) - e \cos(nt)) \\
 y &= -r \sin(t - rr \cos(t + \psi) - e \cos(nt)) \\
 &\quad + r \cos(t - rr \cos(t + \psi) - e \cos(nt))
 \end{aligned}$$

$$\psi = \arctan \left(\frac{\sin((1-n)t) - r \cos((1-n)t)}{\sin((1-n)t) - r \cos((1-n)t)} \right)$$

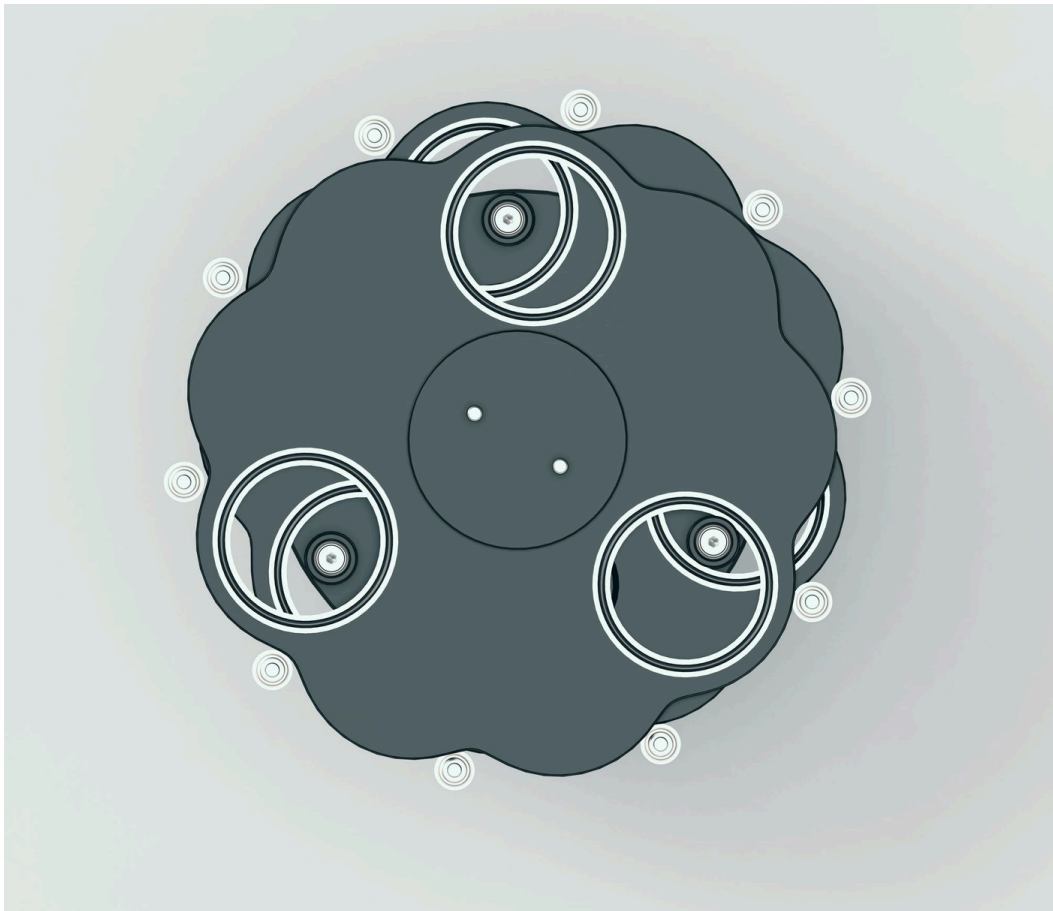
I made an interactive graph on Desmos where you can visualize different cycloid designs quicker than you can in SOLIDWORKS.

▶ 임베드: <https://www.desmos.com/calculator/ivmxq1dkn7?embed>

By plotting these parametric equations from

$$(0 < t < \pi) (0 < r < 1)$$

and mirroring the result I ended up with the final geometry needed for my cycloids.



Design of dual cycloid geometry for the gearbox

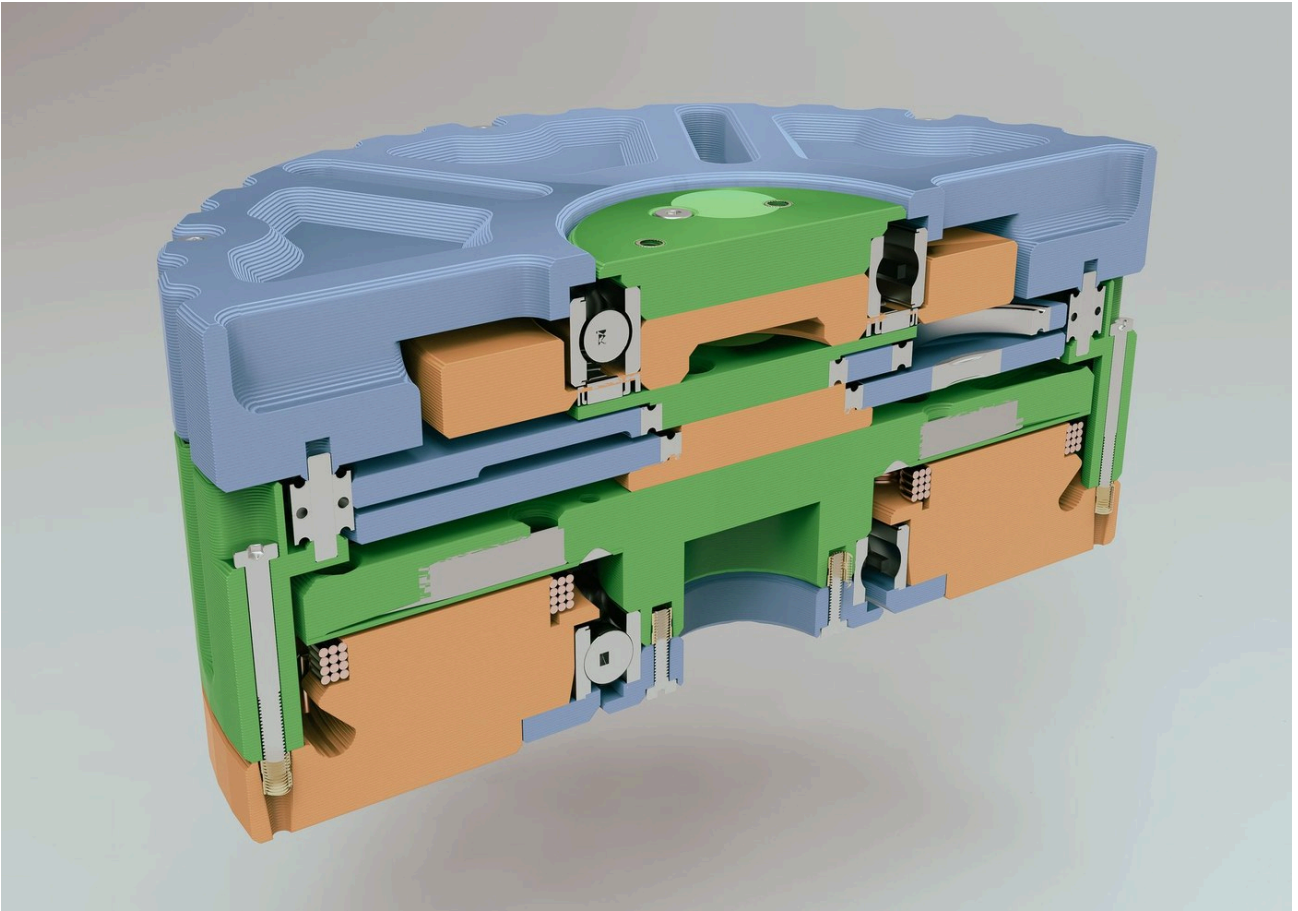
As this design uses two cycloids they actually rotate in the same direction but have inverted eccentricity with the benefit being that a lot of the vibration is canceled out as the center of mass is roughly centered instead of spinning around. This does come at the expense of added complexity.



Cycloid geometry integrated into actuator

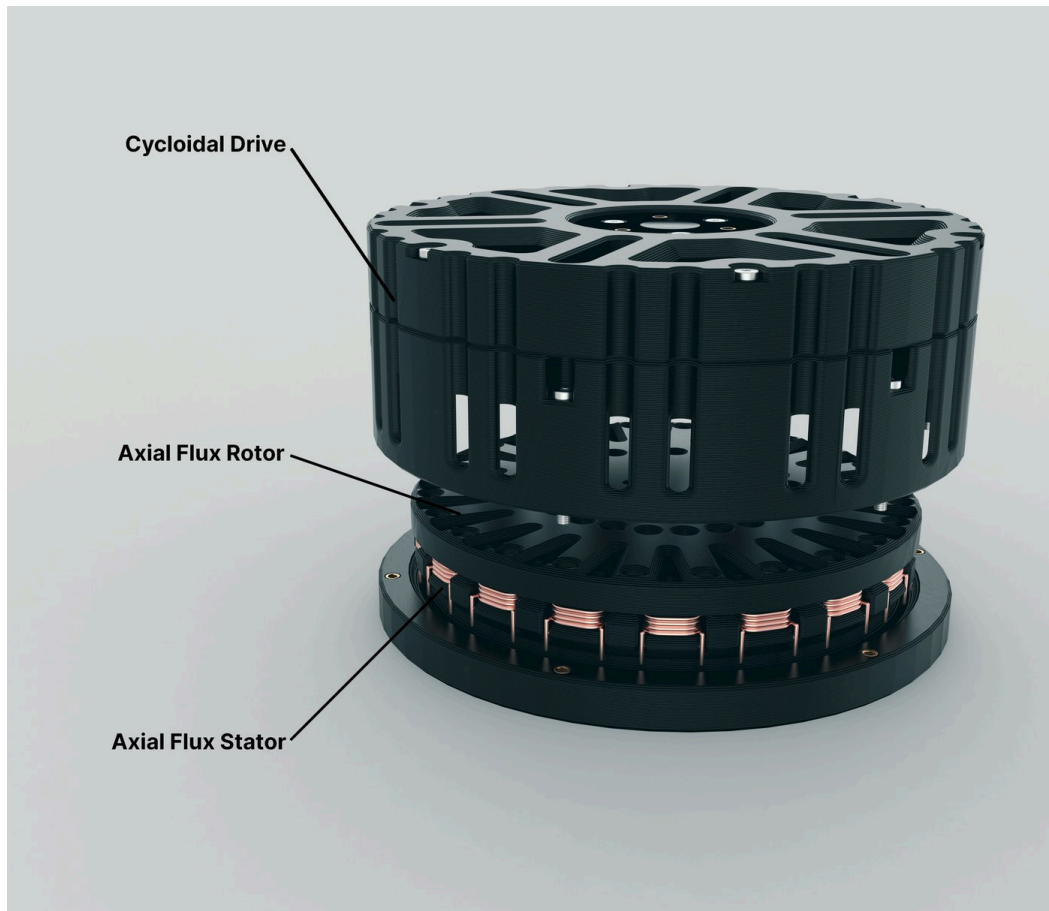
Now it was time to finally integrate everything together into one cohesive design. One challenge I faced during this process was the admittedly arbitrary height limit that I gave myself.

There were a lot of design revisions in order to get all of the parts to fit in the allotted space. Below is an image showing a cross section of the final design with the plastic 3D printed components highlighted.



Section view of the final gearbox design

As you can see above there are some very tight gaps between a lot of the moving parts inside the gearbox as a result of needing to fit in this Z constraint. Below is an image that should give a better idea of how the motor and gearbox were integrated into a single assembly.



Labeled view of the motor and gearbox separated

Once I was done combining the gearbox and motor I landed on this final design

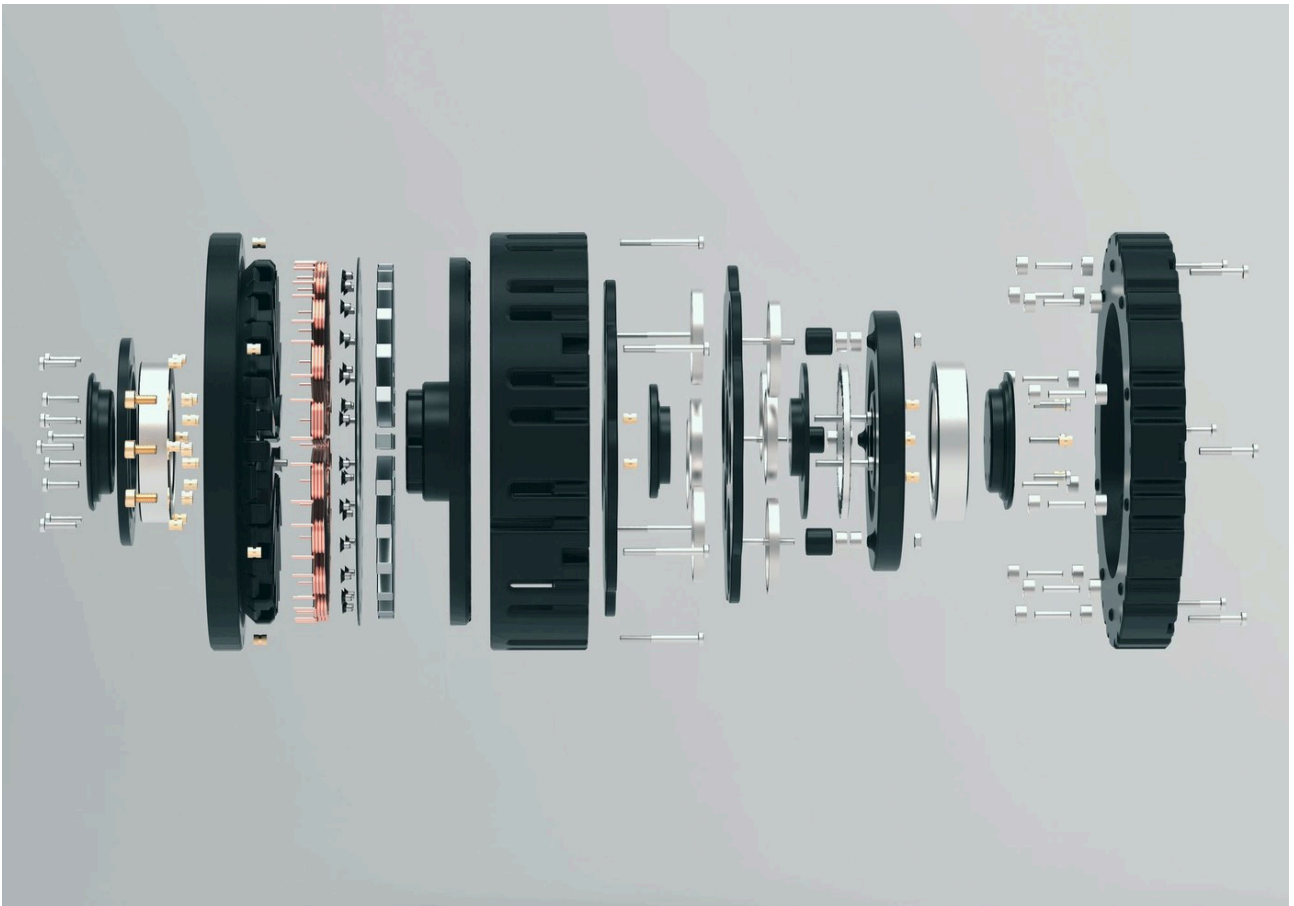
Final Design Renders



Renders of the final design



Explosion view of final design



Horizontal explosion view

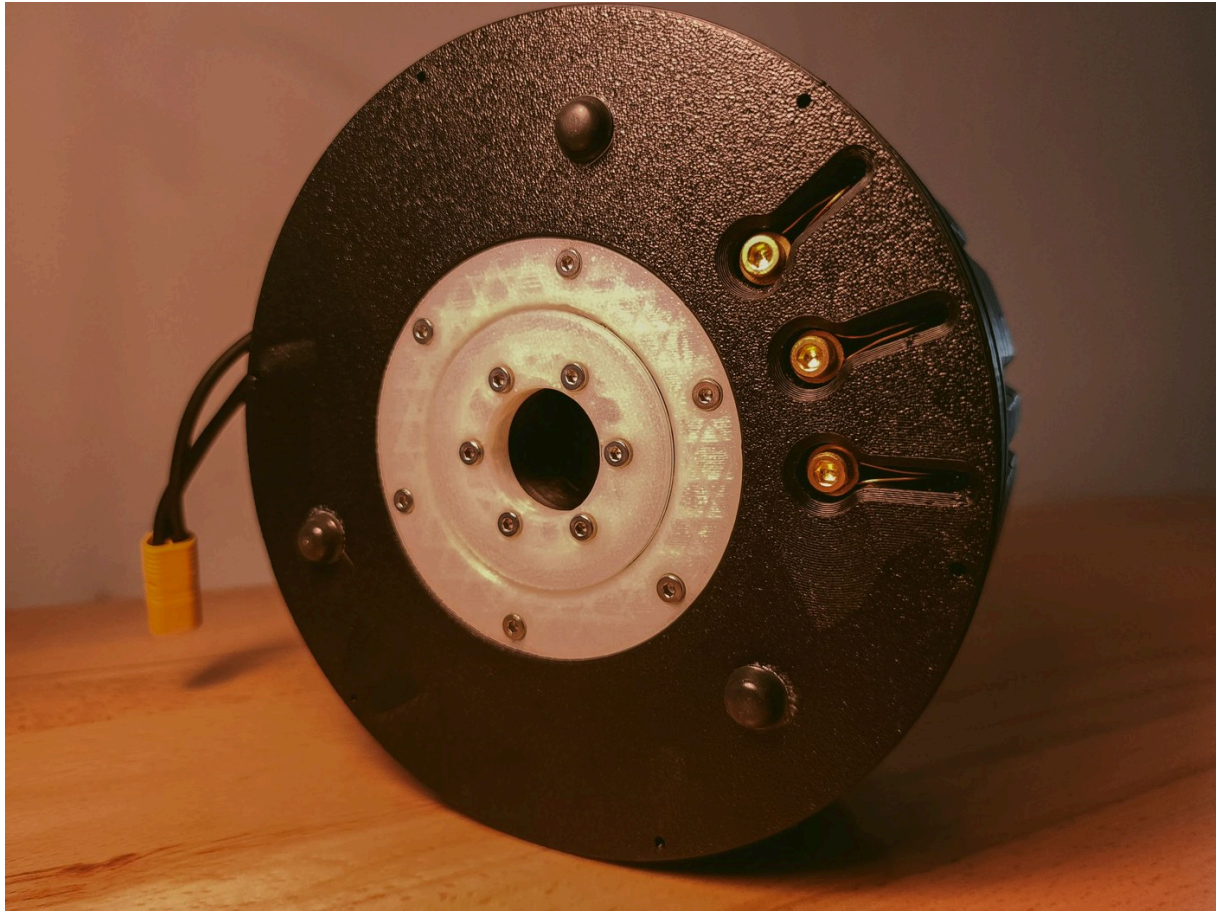
Manufacturing and Assembly

When it came to actually manufacturing the design I had to go through quite a few prototype parts before I landed on the final geometry. There was a lot of friction problems with so many parts tightly packed together moving at high speeds so a lot of incremental improvements had to be made. Below is a picture of all the failed parts that didn't make it into the project.



All the failed prototype parts

However, when I finally did figure out all the issues I was able to assemble and test the actuator. Here are some photos of what the final design looks like assembled.





Different views of the actuator

Testing

Unfortunately I don't currently have access to a dynamometer to do a proper torque test. If I am able to access one I will update this post in the future with the results. Below is a video of me running the actuator on my bench.

▶ 임베드: <https://www.youtube.com/embed/Glw9GQPJTQM?feature=oembed>

I also took a video with the top components removed so you are able to see the cycloids in action.

▶ 임베드: https://www.youtube.com/embed/r_5Vqms3LP4?feature=oembed

My general impressions of it are fairly positive as when it's drawing around ~60W I am unable to stop the output with my hand like I easily could on the previous motor. This combined with the cycloidal nature of the gearbox I cannot feel any backlash whatsoever.

Conclusion

I really enjoyed working on this project as it is a culmination of many of the projects I've posted here over the past few years. Being able to take some wire, magnets, and plastic and assemble them in such a way where they generate rotation with enough torque that even I can't easily stop it is a pretty cool feeling and I am pretty proud of how it turned out.

One area that could be improved is the base torque of the motor. Even just reprinting the stator in Iron PLA would yield a higher torque with zero change to the design.

The other area that this actuator suffers in is friction. With the two cycloids spinning around inside the casing there is more rubbing and friction than I initially intended and I think this could be fixed by either only using a single cycloid or making the actuator taller.

I also wish I was able to do more empirical testing on the motor so I could properly characterize its performance but that may have to be a project for another day.

If you enjoyed this project check out some of my other projects that I have completed below!

Ironless Axial Flux Motor

출처: <https://cadenkraft.com/designing-a-coreless-axial-flux-motor-part-1/>

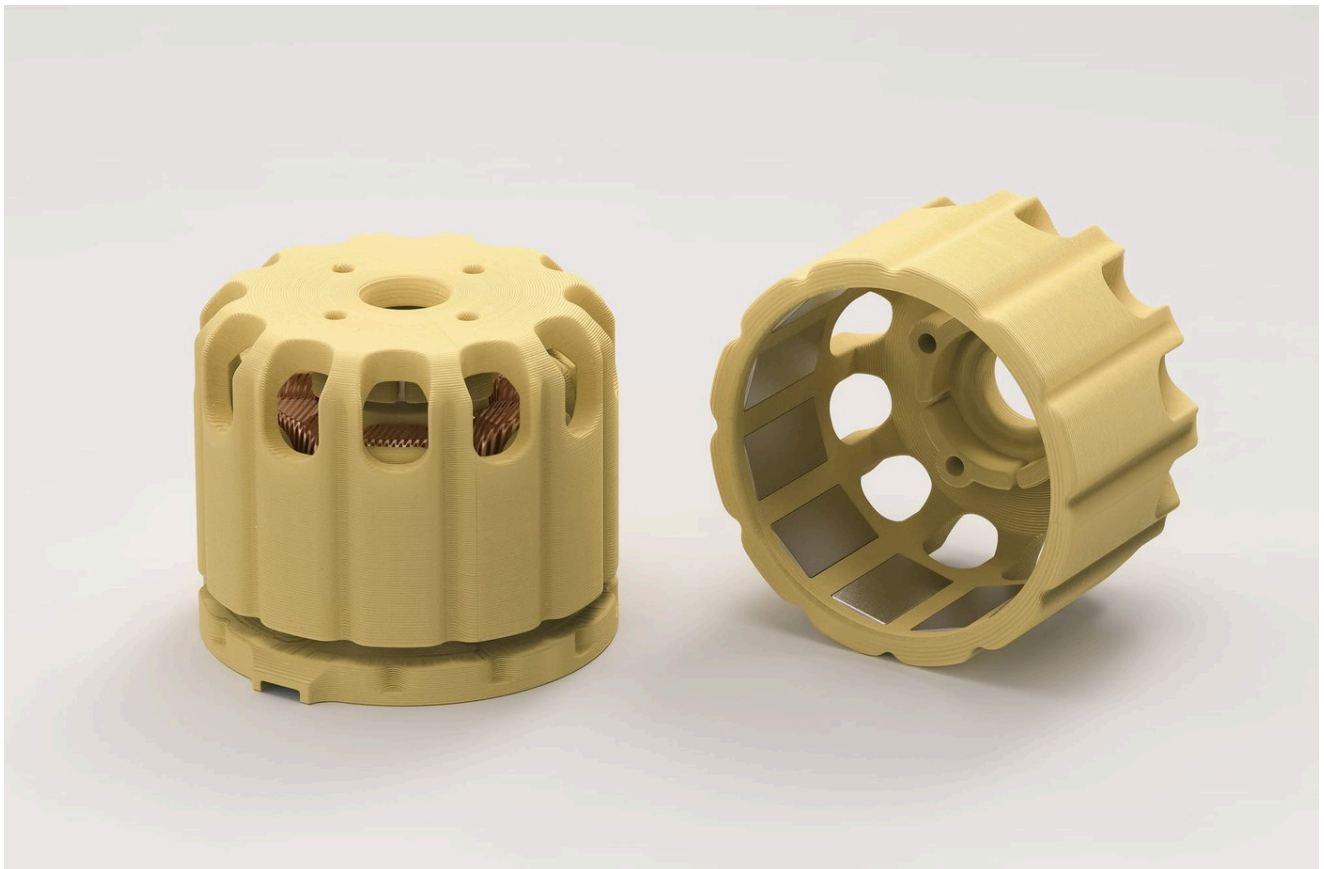
After my first custom brushless motor I felt confident enough with the basic theory of how they work to make a second design with significantly more thought and math defining the design.

Creating a 3D Printed Brushless Motor

This was my first attempt at making a motor from scratch. This design is a simple brushless motor made with components I had on hand at the time and taught me quite a lot about motor design and magnetics.



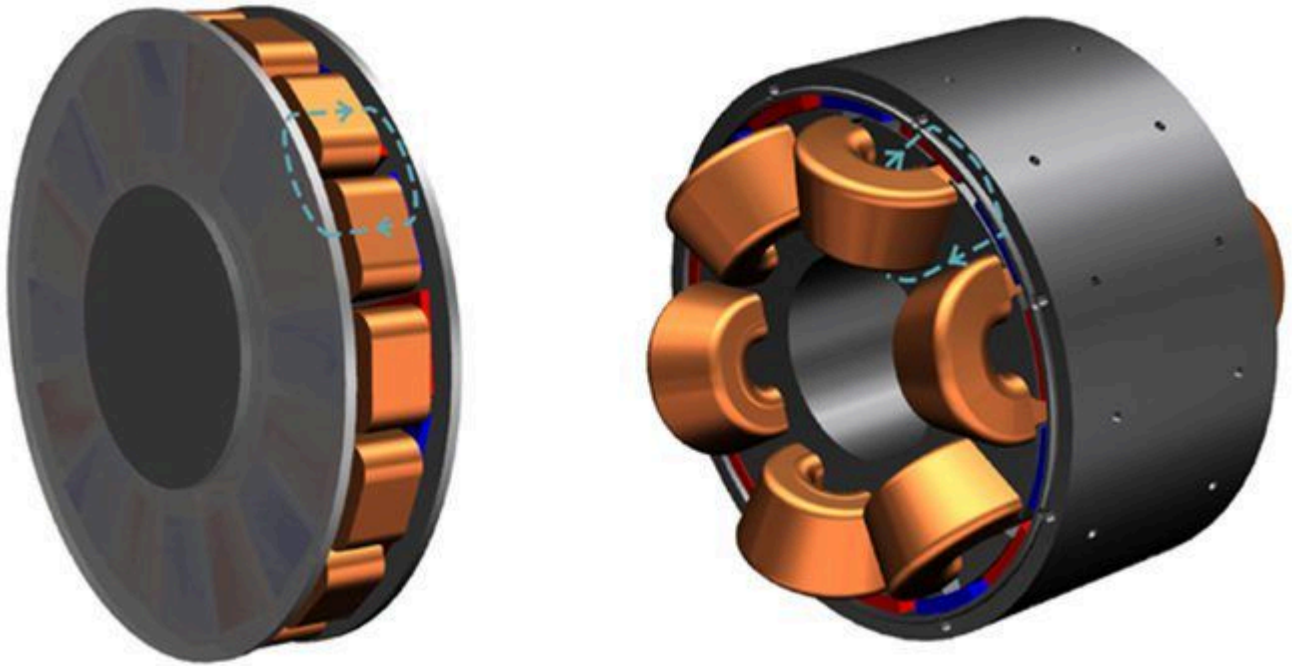
Caden Kraft



As my first design was made by looking at images online and implementing dimensions based on what looked right I found that there were some serious inefficiencies that could be remedied with a bit of math.

Why Axial Flux?

One of the main reasons that I chose to use an axial flux motor design was because the air gap between the magnets and the coils is more or less two dimensional. As opposed to a traditional radial flux motor this allows me to adjust the air gap with shims instead of redesigning the rotor every time to adjust it.



Axial Flux (left), Radial Flux (right)

Another benefit to this design is that because this is a DIY project I do not have the budget to order the custom magnets with a bend for a radial flux rotor.



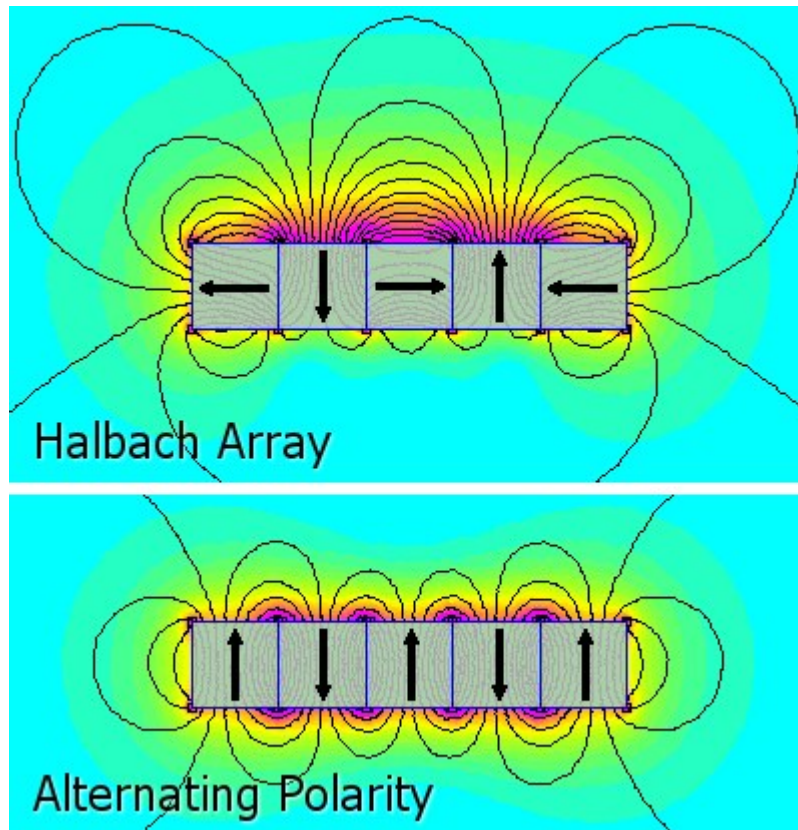
Magnets traditionally used in radial flux motor

The flat nature of an axial flux motor allows me to use traditional rectangular bar magnets that can be found cheaply online in large quantities without the need to have them custom made.

Coreless Design and Halbach Array

Similarly to the magnets, due to cost constraints for this project and the manufacturing capability that I currently have in my college apartment, I am not able to custom make ferrite or steel laminate ferrous cores inside the coils.

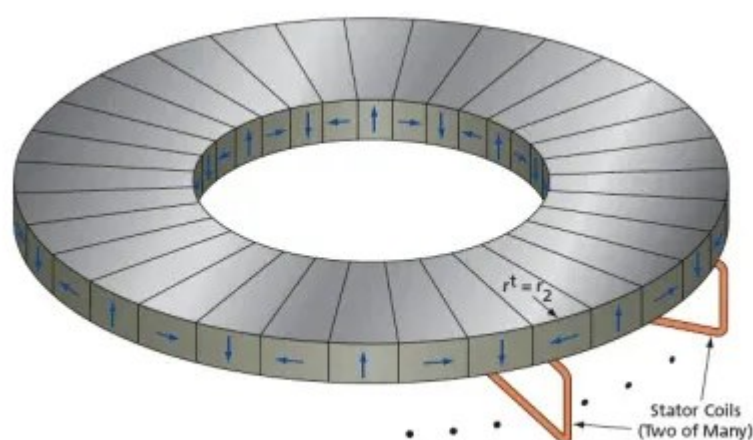
As a ferrous core is extremely important in shaping the magnetic field around the coils, I had to research a different solution to remedy this and came across the Halbach magnet array. This works by alternating the magnet poles by 90 degrees at a time instead of 180 degrees like normal designs.



Halbach Array (top), Alternating Array (bottom)

This uses extra magnets to shape the magnetic field as a iron core traditionally would. As there are very little designs on the internet that I could find the utilize a Halbach pattern I chose to implement it into my design to hopefully contribute some experience implementing it into a motor once the design is completed.

This means that while the rotor of my motor has 12 poles it actually utilizes 24 magnets to implement the Halbach design.



Example of Halbach array pattern

Defining Specifications

Next I had to define a few specifications around various aspects of the motor needed to calculate the number of coil turns needed for the final design.

I found an excellent research paper online that outlined the mathematical parameters needed to successfully create a Halbach axial flux design.

[PDF] Paper 088 , ENG-xxx Design and Test of an Ironless Axial Flux Permanent Magnet Machine using Halbach Array | Semantic Scholar

Conventional permanent magnet electric machines utilize ferromagnetic cores of high permeability. The high core permeability relative to the surrounding material helps to concentrate the magnetic flux in the core, and guide the flux across the airgap of the machine. An ironless machine design remove...

Semantic ScholarSemantic Scholar's Logo

T. Batzel

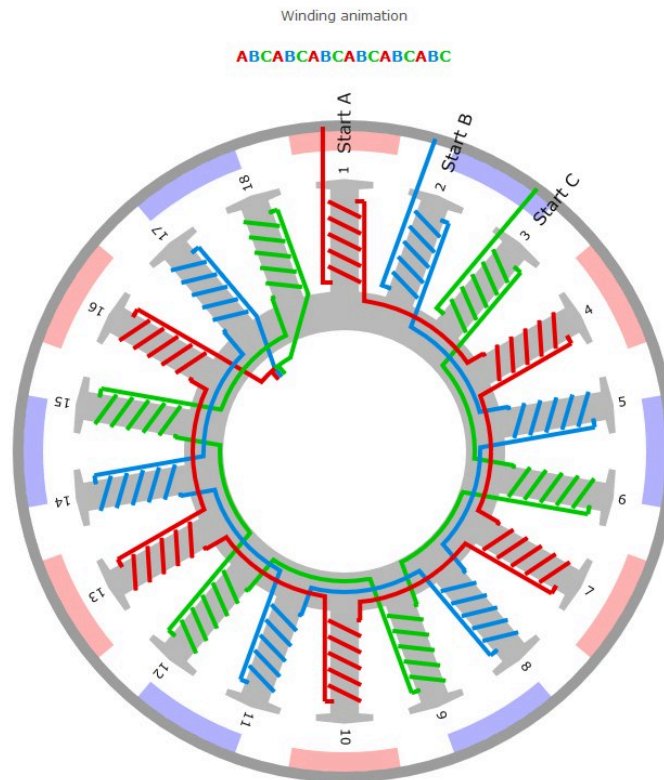
Design and Test of an Ironless Axial Flux Permanent Magnet Machine using Halbach Array

Through this paper and some preliminary CAD design as well as a quick inventory of all of the motor controllers and bearings that I had already, I was able to define certain parameters needed to calculate the coil turns below. The coil area was calculated by taking the approximate area enveloped by the outer perimeter of copper of a single coil.

Parameter	Value
DC Bus Voltage	24 V
Base Speed	700 rpm
Rated Current	7 A (rms)
Wire Diameter	0.65 mm
Poles (p)	12
Coils (c)	18
Connection	Star (Y connection)
Coil Area (A _{coil})	$4.92 \cdot 10^{-4} \text{ m}^2$
Rotor Area (A _{rotor})	$9.42 \cdot 10^{-3} \text{ m}^2$
Outer Radius (r _o)	65 mm
Inner Radius (r _i)	35 mm
Magnetic Flux Density (B)	0.5 Tesla

I chose for the motor to have 12 poles and 18 coils as it has an efficient high winding factor.

This slot/magnet pole combination will have **36** cogging steps per turn. $LCM(18,12)$
And its winding factor is: **0.86603** || advanced



Winding Factor from Bavaria Direct Online Calculator

Calculating Parameters

The first step in defining the coil turns needed is calculating the area of the motor that is covered by the magnets on the rotor.

A
rotor
= $(p \cdot 2) \cdot ($
0.036
m
0.006
m)
rotor
= $($
 $\cdot 2) \cdot ($
0.036
 $\cdot$
0.006
)

Next, I needed to define an optimal outer to inner radius ratio in order to maximize the efficiency of the motor. As the paper above says, an optimal ratio for this is around 0.5. This allowed me to define the inner and outer radius based on the 36mm*6mm*6mm magnets that I found online.

$$\alpha = r_i \omega$$

=

Then I had to calculate the peak flux density using the equation below as well as the recommended mean air gap flux density of 0.5 Tesla from the paper above.

$$\tau = F_{\text{rotor}} = B l_i l_w = B i_w = B Z$$

=

=

=

=

$$B = 2 B_m \pi$$

$$= 2$$

After using these two equations I arrived at a mean air gap flux density of 0.785 Tesla. Using the next couple equations I was able to use this value combined with others above to figure out the final number of coil turns.

$$e_{ph} = N_{ph} A_{coil} \frac{dB(\theta)}{dt}$$

$$h =$$

h

(

)

$$B(\theta) = B_m$$

$$\sin$$

$$(\omega t)$$

(

$$)=$$

$$\sin$$

(

)

$$e_{ph} = N_{ph} A_{coil} \omega B_m$$

$$\cos$$

$$(\omega t)$$

$$h =$$

h

$$\cos$$

(

)

$$N_{ph} = \frac{e_{ph}}{A_{coil} \omega B_m}$$

$$h =$$

h

The electrical frequency of the stator voltage can be determined using the shaft speed and number of pole pairs.

$$\omega_e =$$

$$700$$

$$\frac{rpm}{60} \cdot 2\pi \cdot 60$$

$$= \frac{700}{60} \cdot 2 \cdot 6$$

Plugging in the final numbers into the equation below equates to about ~87 coils per phase of the motor.

$$N_{ph} = \frac{24}{V\sqrt{3}} \cdot \frac{4.92}{10} \cdot \frac{700}{rpm} \cdot \frac{60}{s} \cdot 2\pi \cdot 60 \cdot 0.785 \cdot T$$

$$h = \frac{24}{3} \cdot \frac{4.92}{10} \cdot \frac{700}{60} \cdot \frac{2}{6} \cdot 0.785$$

$$N_{ph} = 81.57$$

$$h = 81.57$$

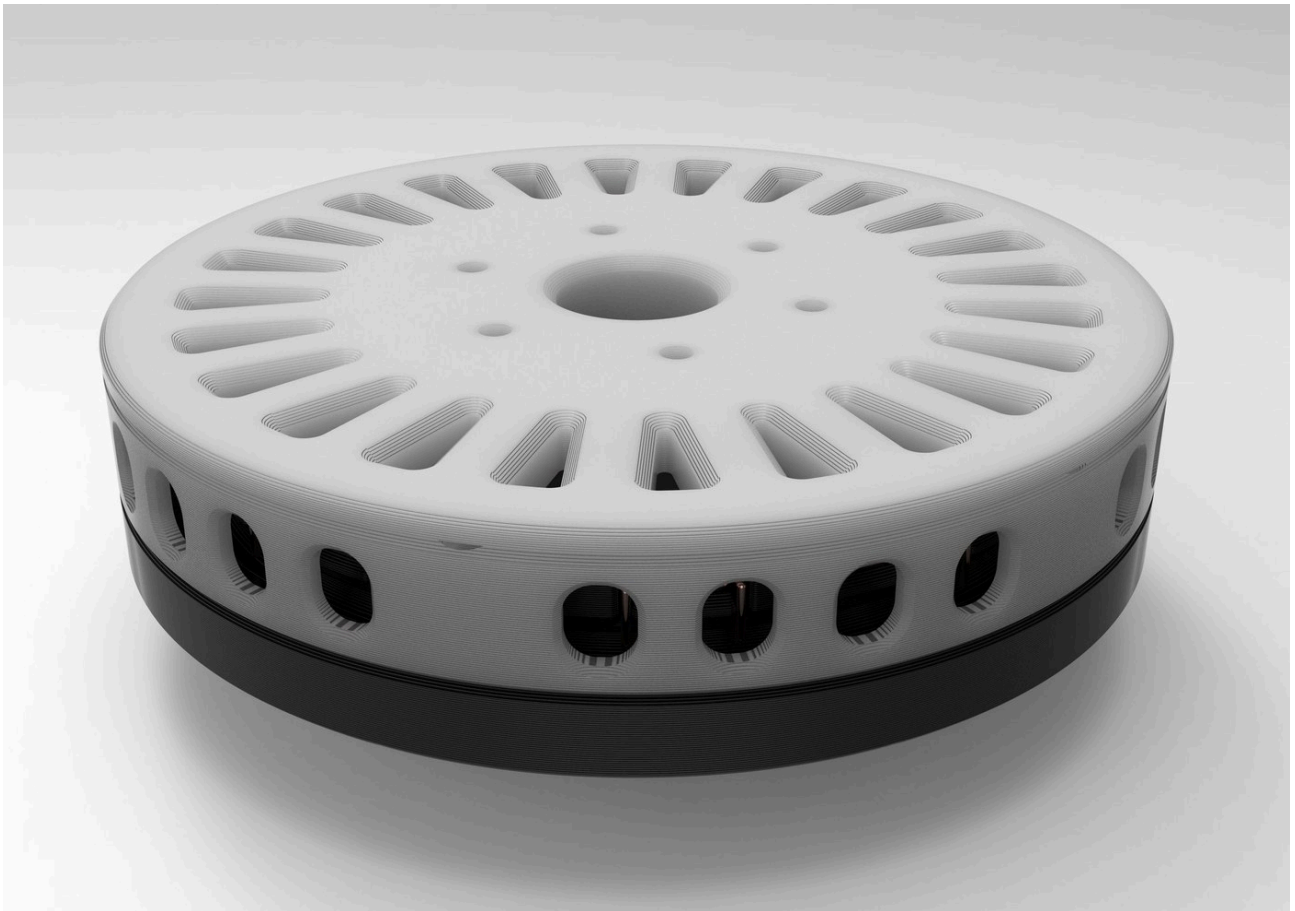
Dividing this by six as there are 6 coils per phase it equates to around 14 turns per coil.

Finalizing the Design

Now that I had the parameters defined I started work on creating the final design. The final design will consist of mostly 3D printed parts and a 62mm*40mm*12mm unsealed roller bearing. This

bearing has significantly more rigidity than the 608 skateboard bearing I used in my first design.

Here are a few renders of the final motor design that I decided on. The white/clear parts are printed in PLA while the base stator is made of PETG for the extra temperature resistance as the coils heat up.



Completed assembly

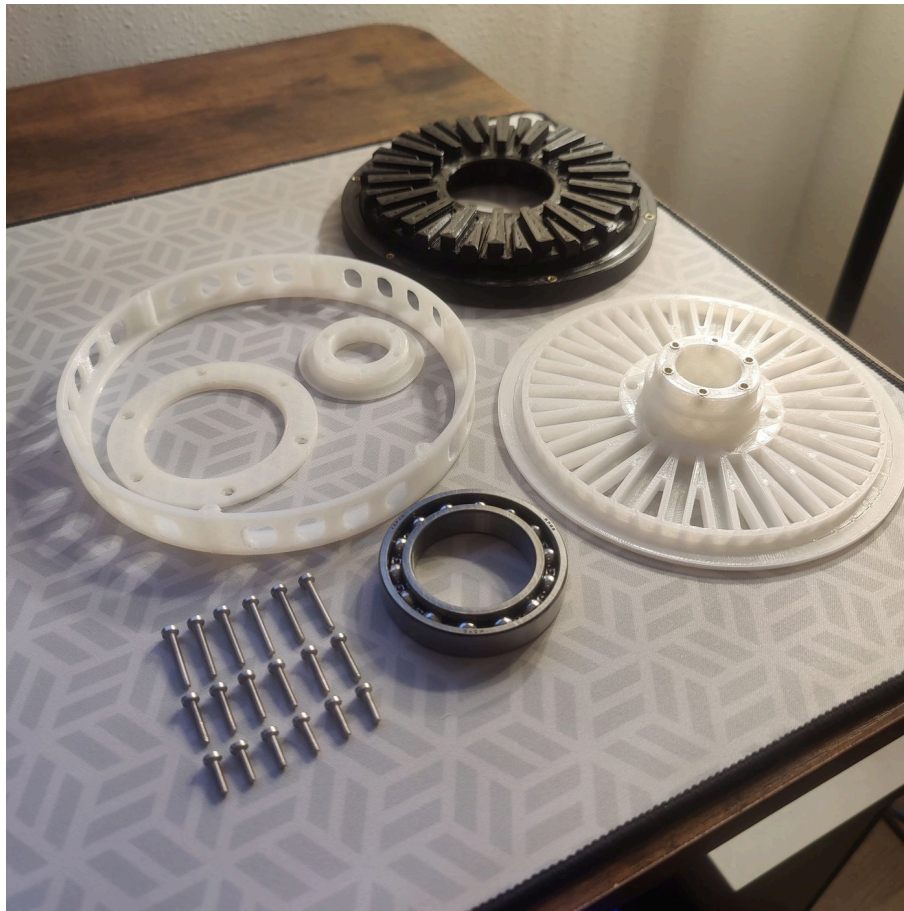


View of the magnets and coils with the rotor hidden

Below is a draggable render where you can see the final explosion view of the design from multiple angles.

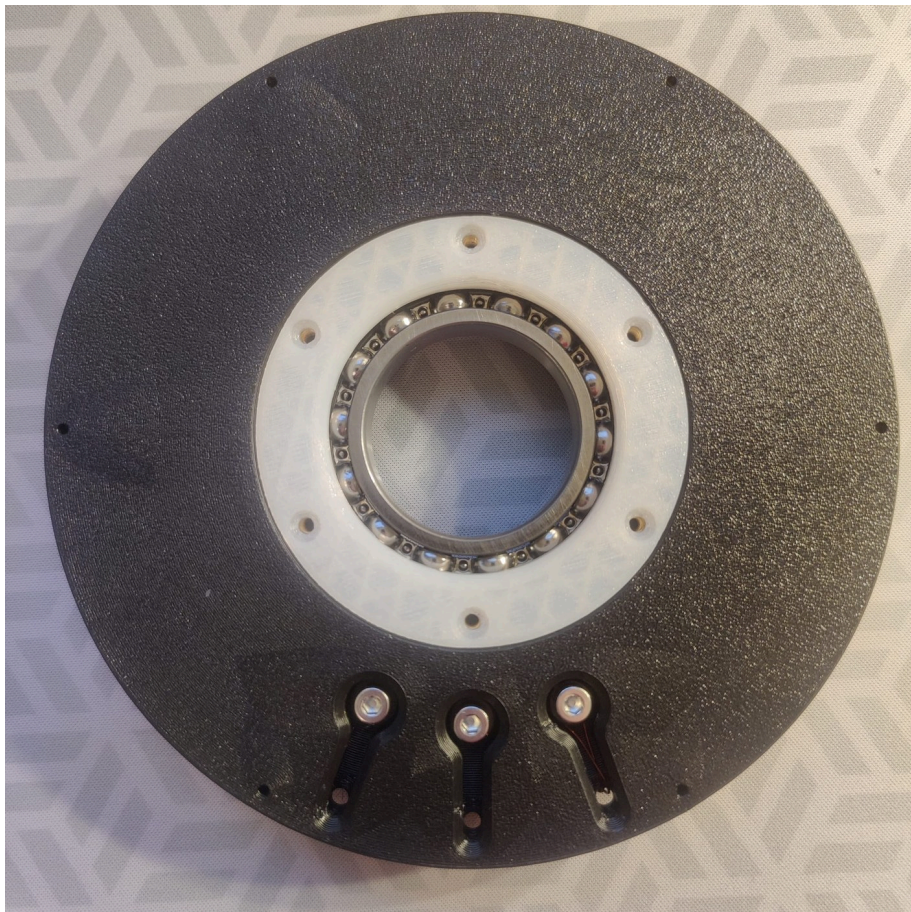
▶ 임베드: https://cadenkraft.com/content/renders/axial_flux_animation.html

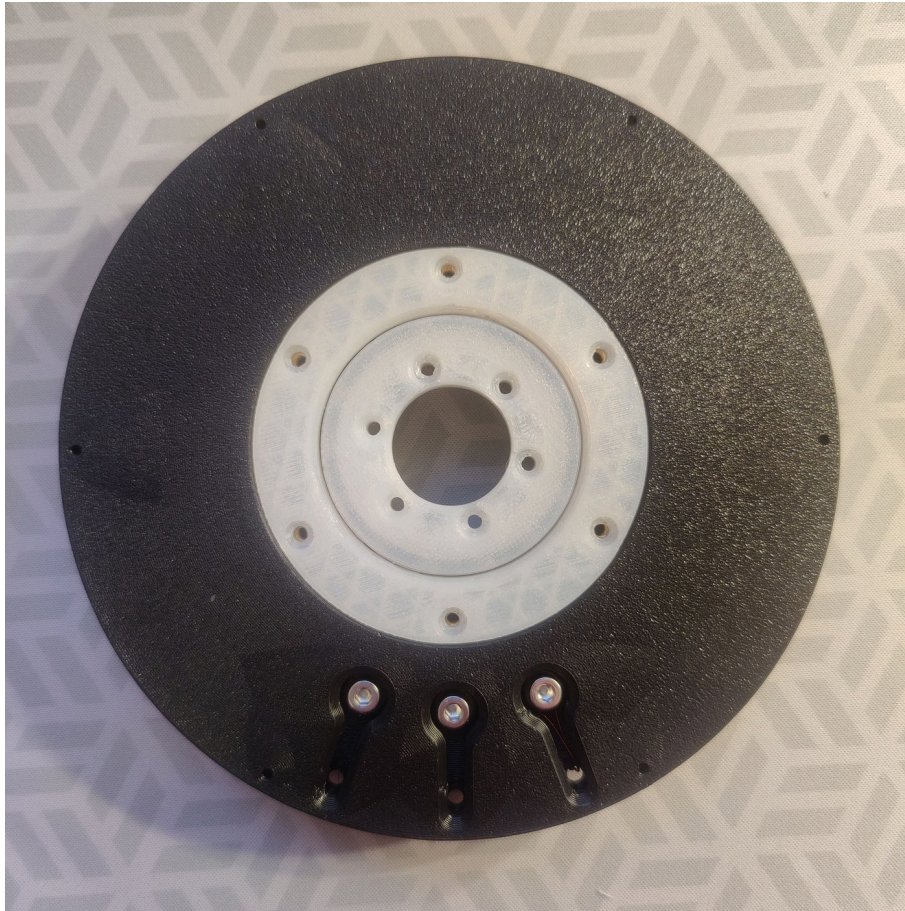
Here is an image of all of the different printed parts and hardware (aside from magnets and enamel wire) needed to assemble the motor.



All completed printed parts

These photos show a more in depth look at the design that I used in order to secure the bearing to the plastic piece. This has proved to work significantly better than the friction fit bearing mount seen in my first design.





Pictures of the parts I used to secure the bearing to the rotor and the stator

Finally here is an image of the final design assembled (again without magnets or coils) with a Sharpie for scale.



Last but not least, I thought I would throw in an image showing the amount of prototype parts it took to get the fitment of everything just perfect.



Assembly

Now that the printed parts are completed and they are successfully test fit I ordered the necessary magnets and wire needed to complete the motor and finally get it up and running.

The step-by-step plans for what the assembly will include are listed as below.

- Wind all 18 coils coils (0.65mm Enamel Wire)
- Measure coil resistance with University LCR meter
- Solder power leads and connectors to coils
- Assemble magnets into rotor (36x6x6mm Neodymium magnets)
- Create test stand to power and characterize properties of the motor

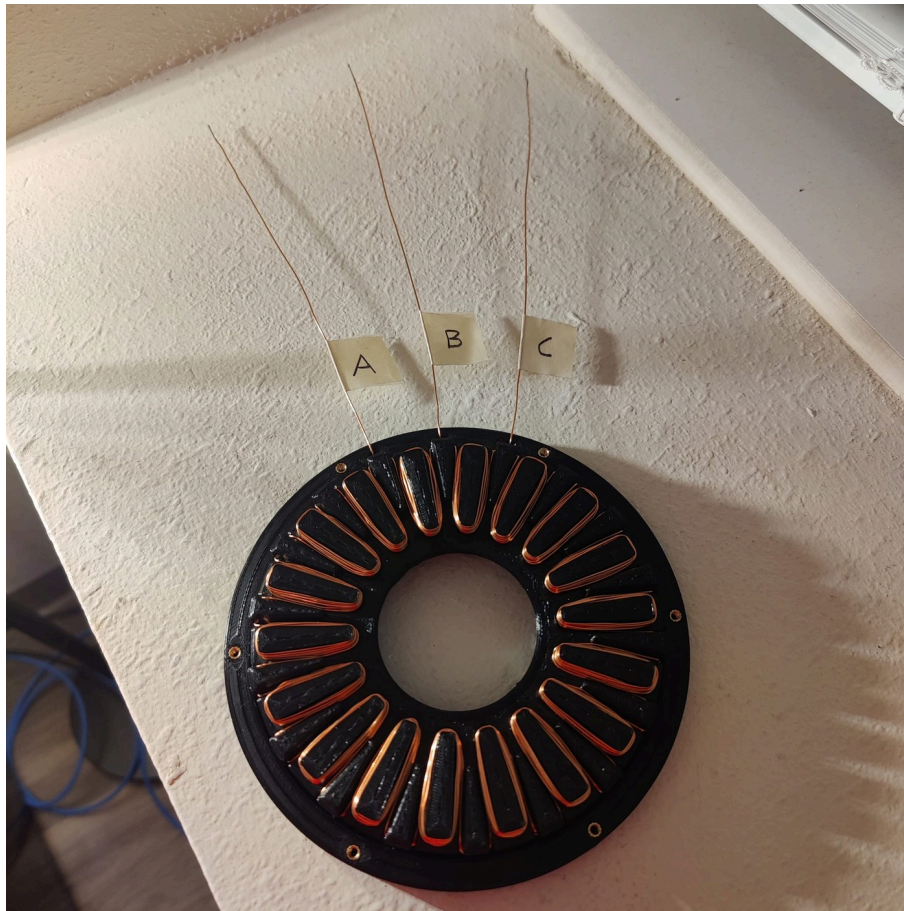
Here are the photos of the wound coils for the motor as well as the magnets in the rotor. I used a magnetometer app to find the northern pole of the magnets in order to properly construct the Halbach array.





Magnets in rotor (left), Would coils (right)

In order to make sure that I wound the coils correctly and the solder joints didn't induce any added resistance I took the stator assembly to one of the university LCR meters to measure resistance and inductance.



Labeled coils (left), LCR meter setup (right)

Below is a table summarizing the LCR measurements.

	AB	AC	BC
Resistance (mOhms)	699	705	708
Inductance (uH)	56.45	57.69	57.19

While coil A differed slightly I deemed this within my allotted margin of error and moved forward. Finally once it was fully assembled I was able to hook everything up and hope that it would spin. Below is a video of its first spin.

▶ 임베드: <https://www.youtube.com/embed/3goDI7w70WA?feature=oembed>

Video of me spinning up the motor for the first time.

I would have liked to test the torque of the motor with a dynamometer but I did not have access to one at the time. Maybe I will save this for a future project but for now I'll have to leave this one here.

Conclusion

I really enjoyed this project because it was the first project I've done that required me to do a lot of research and get a baseline understanding of what was needed to complete the project before even starting.

This took quite a long time to make but I think it was ultimately worth it as I have vastly expanded my knowledge of motors and motor design and I actually was able to create a working motor. Hopefully I am able to carry the skills that I learned through this project onward to future ones.

Thanks for reading this post. If you are interested in any of my other posts they are linked below!

References

[1] T. Batzel, A. Skraba, and R. D. Massi, "Design and Test of an Ironless Axial Flux Permanent Magnet Machine using Halbach Array," IAJC-ISAM Joint International Conference, vol. 88, 2014.